

View: agents will integrate real digital tools to solve real tasks; agent scaling raises the capability ceiling; self-optimising agents accelerate AI iteration; agents create new training/inference infra challenges and require immediate investment

Trend 1, model: in 2H25 the key to agentic models is combining multimodality, slow thinking and tool calling into a compound capability

Trend 2, form: agents become stateful, long-task and event-driven; interaction shifts from synchronous Q&A to asynchronous proactive recommendation plus user selection

Trend 3, application: agents self-optimize infra and accelerate AI iteration — build agents to build infra for AI; the product expression is the coding agent

Trend 4, technology: agent scaling spans environment, agents, long trajectories, rewards and memory; reward scaling is critical

Trend 5, infra: multi-programme, multi-stage, multi-model, multimodal and multi-tool workloads require agentic RL frameworks to support asynchronous RL, separated deployment and efficient resource scheduling to improve end-to-end throughput

Trend 6, evaluation: agent evaluation differs from model evaluation; industry tests agent capability in real scenarios across task complexity, precision, reliability and completion efficiency

Overall trend of agentic models

Agent definition: an intelligent system that uses digital tools to complete real-world tasks

OpenAI: reasoning, tools and long context

Agents = Reasoning + Tools + Long Context
Tools: browsing / search, computer use, canvas, etc.

The future: innovator = agent + creativity, with human-agent collaboration

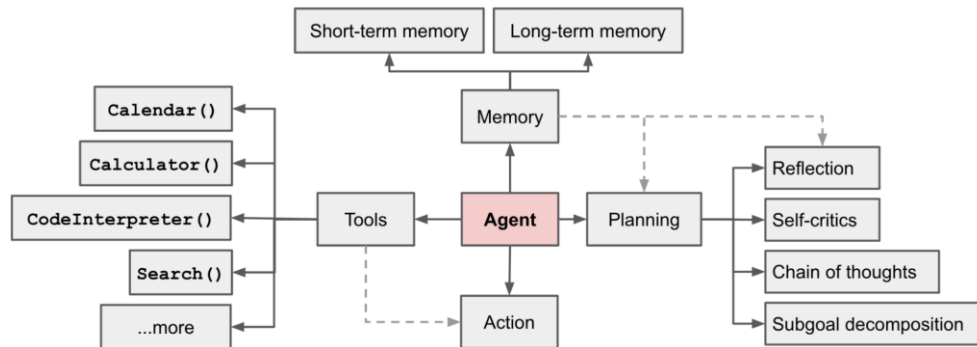
Co-Innovators = Agents + Creativity (human-AI collaboration)

Anthropic criteria for using agents: complex, high-value, feasible, and low cost of error

Checklist: "Should I build an agent"

Is the task complex enough?	No → Workflows Yes → Agents
Is the task valuable enough?	<\$0.1 → Workflows >\$1 → Agents
Are all parts of the task doable?	No → Reduce scope Yes → Agents
What is the cost of error/error discovery?	High → Read-only/human-in-the-loop Low → Agents

Agent = action + planning + memory + tool use



Agent experience

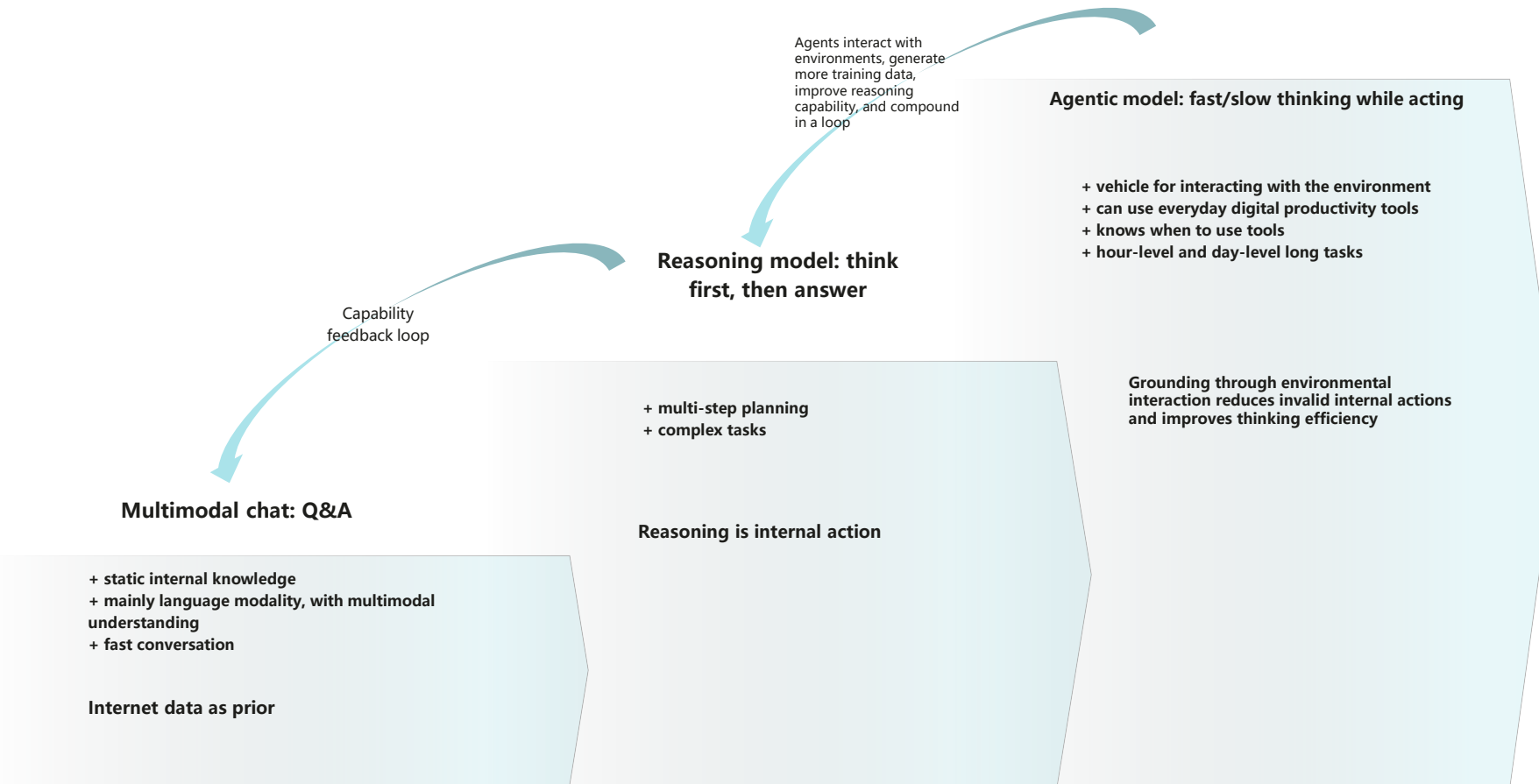
Task complexity ~ time required

Completion efficiency ~ value minus cost

Completion accuracy ~ pass@k;

Completion robustness ~ pass^k, still lacking

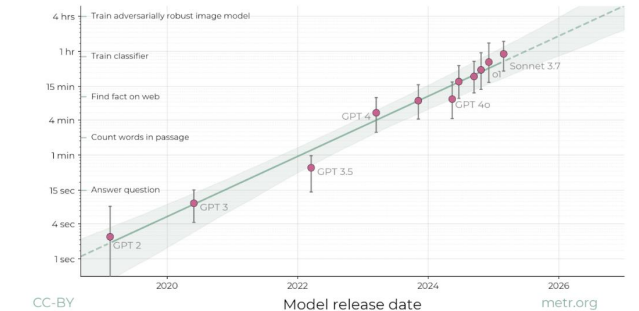
Why agents: the shift from Q&A to action; using real tools to solve real problems. Agents are the next platform layer and also feed back into reasoning.



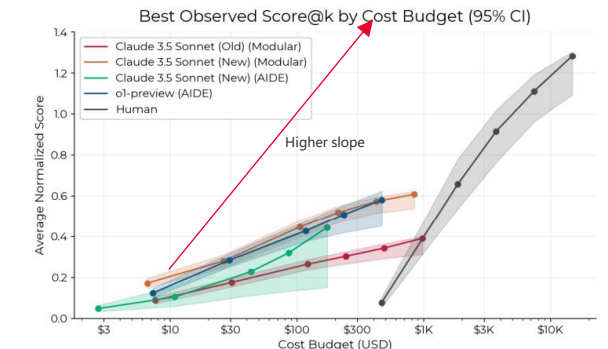
Solve real-world problems

Replace or augment real digital productivity tools with agents and directly create economic value

The length of tasks AI can do is doubling every 7 months
Task length (at 50% success rate)



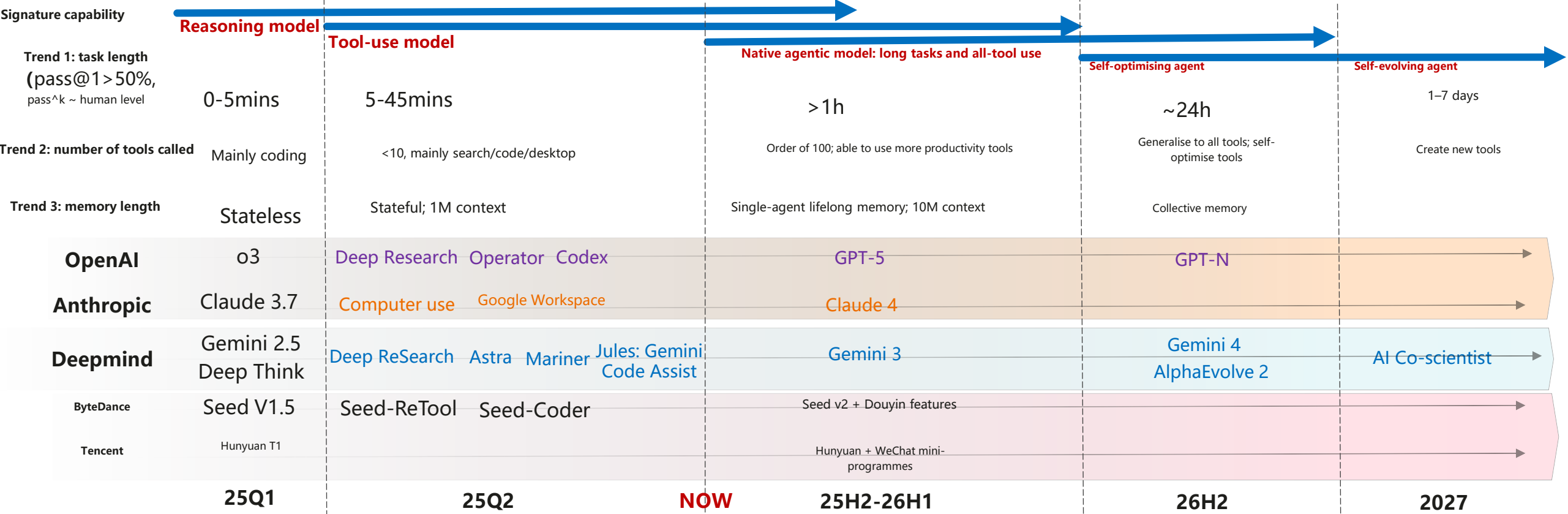
Can reliably complete long tasks in the future



Future systems must exceed human marginal output efficiency

Agentic capability iteration is accelerating; models already have capability in specific easily verifiable tools. Agentic models may break out in 2H25 with

commercial value; leaders will compete on full-stack integration and iteration speed, with important applications emerging in 1H26.



2025 model iteration is accelerating: RL opens the data flywheel, shortening cycles from 6-12 months to 3-6 months. In 2H25, end-to-end training is expected to produce the first native agentic models with multimodal thinking and autonomous tool calling. In 1H26, more real digital tools should become callable by models; multimodality + thinking + coding will compound into a capability step-change. With user-history data, personalised always-on general assistants may appear.

- Model advantage, OpenAI / Anthropic: use stronger models to launch agents in 2H25 and improve coding capability; by year-end, a single model/product may integrate multiple tool capabilities; in 2026, richer interactive environments help generalise agentic capability.
- Internet advantage, Tencent / Meta: agentic RL scaling needs fast interaction with real environments and scalable reward systems. The internet is a strong interactive environment and enables the super-app advantage.
- Combined advantage, Google / ByteDance: strong models plus internet-environment advantages can produce agent forms around their own products, accelerate RL through user-data flywheels, and speed model-product iteration. A prototype may be visible before 2H26.

2H26: mature agentic models may complete real tasks and accelerate actual work. Companies with more advanced agentic models will iterate faster, developing next-generation models, compute infra and products faster: build AI to accelerate AI infra for AI. We may see an arms race in self-evolving agents, with laggards falling further behind.

AI infra: as RL workload complexity rises, multi-programme + multi-stage + multi-model + multimodal + multi-tool characteristics require RL frameworks to support asynchronous RL and efficient system scheduling for agentic RL training and higher end-to-end throughput. The industry is extending frameworks such as veRL and skyRL. Gaps between RL frameworks can be 100-1000x, and better frameworks will support faster model iteration.

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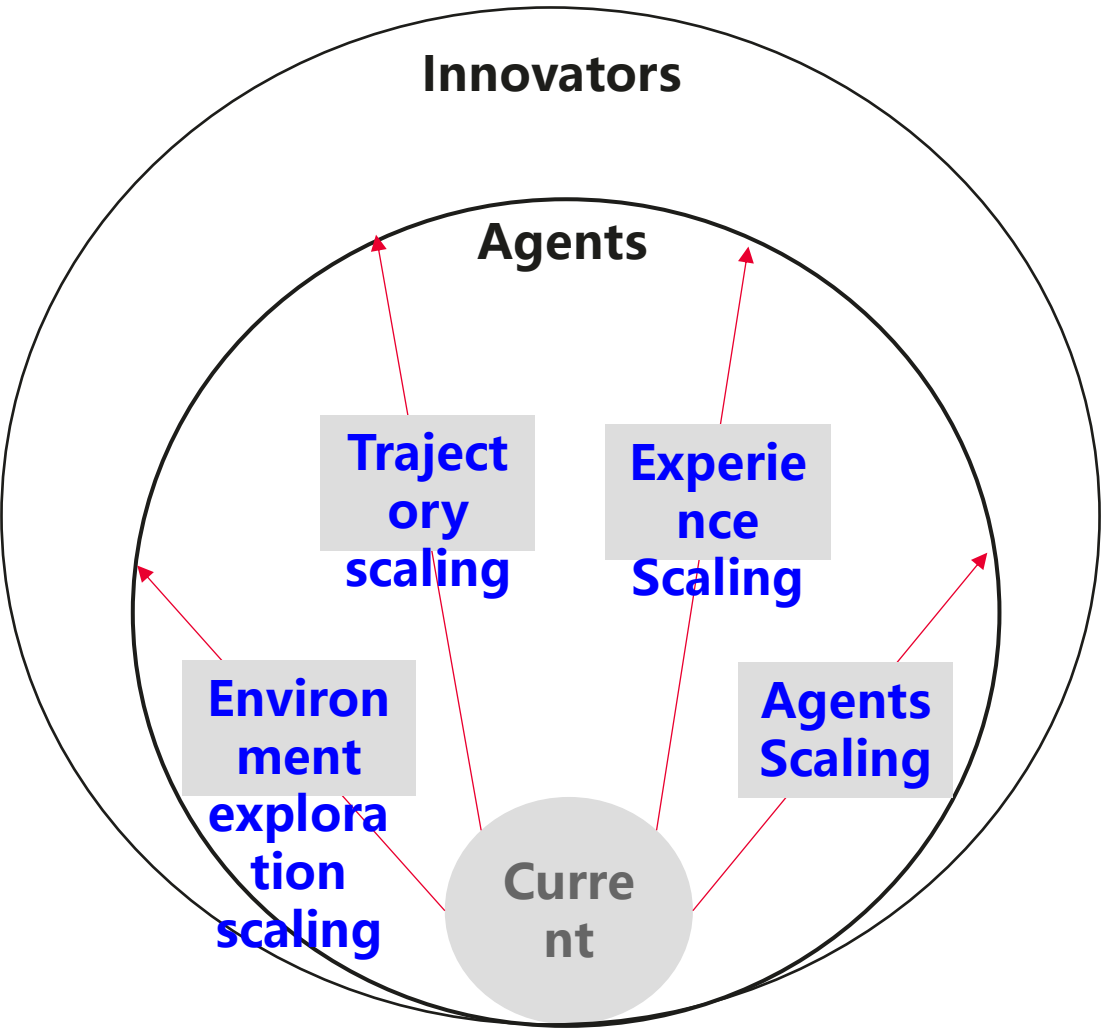
commercial value; leaders will compete on full-stack integration and iteration speed, with important applications emerging in 1H26.

Signature capability	Reasoning model	Tool-use model	Native agentic model: long tasks and all-tool use		Self-optimising agent	Self-evolving agent
Trend 1: task length (pass@1 > 50%, pass^k ~ human level)	0-5mins	5-45mins	>1h		~24h	1-7 days
Trend 2: number of tools called	Mainly coding	<10, mainly search/code/desktop	Order of 100; able to use more productivity tools		Generalise to all tools; self-optimize tools	Create new tools
Trend 3: memory length	Stateless	Stateful; 1M context	Single-agent lifelong memory; 10M context		Collective memory	
OpenAI	o3	Deep Research Operator Codex	GPT-5		GPT-N	
Anthropic	Claude 3.7	Computer use Google Workspace	Claude 4			
Deepmind	Gemini 2.5 Deep Think	Deep ReSearch Astra Mariner Jules: Gemini Code Assist	Gemini 3		Gemini 4 AlphaEvolve 2	AI Co-scientist
ByteDance	Seed V1.5	Seed-ReTool Seed-Coder	Seed v2 + Douyin features			
	25Q1	25Q2	NOW	25H2-26H1	26H2	2027

System	+ asynchronous agentic RL training/inference			The future of the Gemini models	
Base model				+ native video generation	
RL algorithm	+ RLVR			+ new algorithm for autonomous online learning?	
reward	+ rule based+GRM + reward scaling			+ intrinsic reward	
Environment	+ web, compiler			+ internet, software, apps	
Long sequence / memory				+ world model • Big models Scale is all you need, we know	
				+ infinite context	

- Omnimodal (r)
Already have image + audio generative natively, next is video
- Early experiments on Diffusion (r)
- Agentic by default (m)
Research class tool calling and tool use, but more than that, models are becoming agents
- Reasoning scaling will continue (s)
Research breakthrough after research breakthrough
- More small models (s)
More to share here soon!
- Infinite context (r)
The way attention and context works today, this is never going to be possible. We need new innovation at the core architecture level to enable this

Agentic AI evolution: new scaling modes drive rapid progress in agentic models

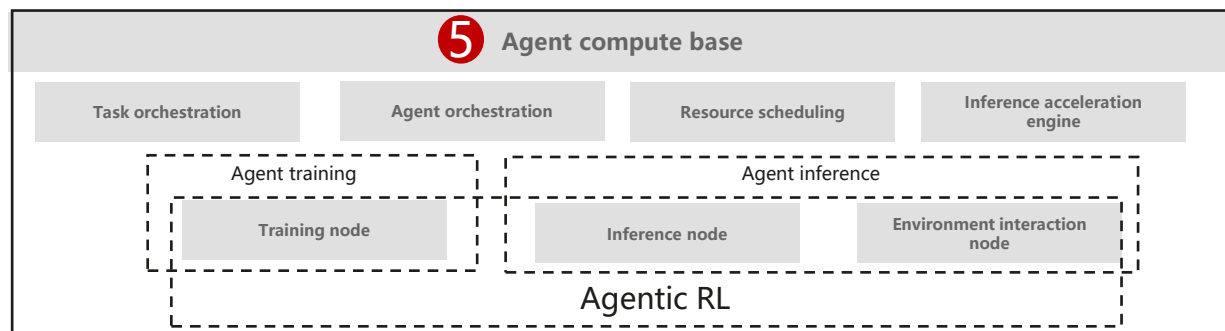
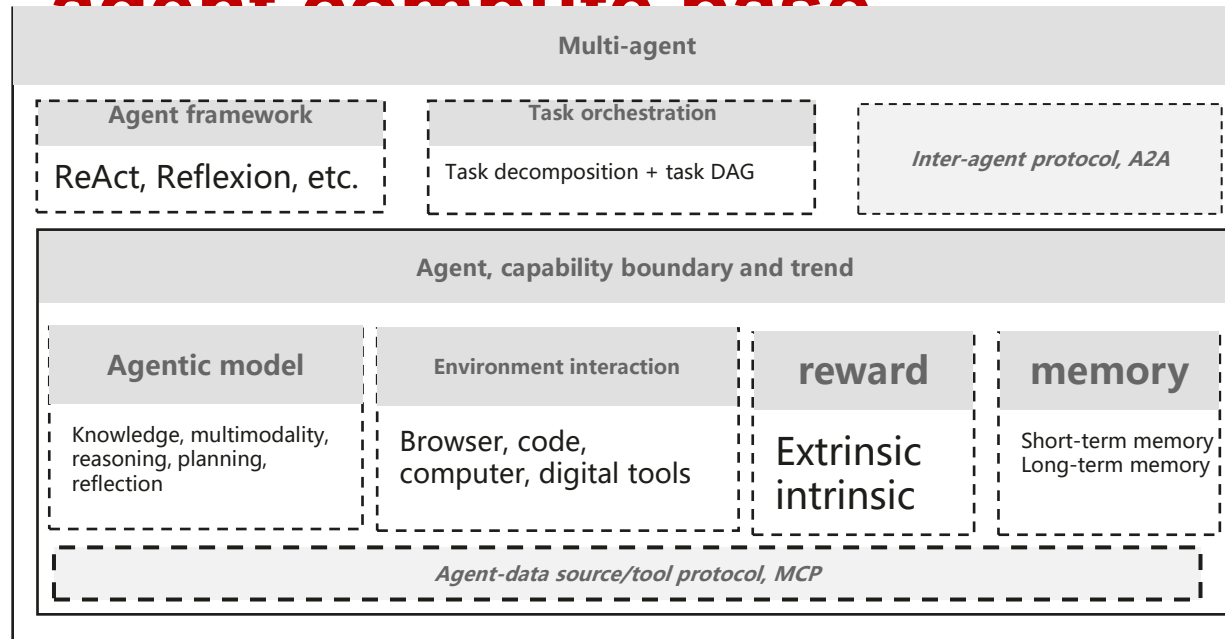


Agentic AI remains early. Environment-exploration scaling, trajectory scaling, experience scaling and agents scaling will materially raise capability.

Core trends

1. Open-environment exploration: move from single-environment exploration to joint exploration across complex environments. Open rewards become critical, and world models can act as environment simulators.
2. Ultra-long trajectory learning: task duration doubles every seven months. Long-task capability depends on long-trajectory learning; long-trajectory data, rewards, training and memory all remain challenging.
3. Experience scaling, self-evolution: expanding the exploration space, creating infinite experience and iterating internal understanding is the core loop for autonomous evolution.
4. Collective intelligence: multi-agent systems have advantages in parallel efficiency and multi-path exploration. Jointly optimising collaboration raises exploration capability and enables socialisation inside agent networks.
5. Compute-base impact: for multi-environment interaction, long trajectories and multimodality, systems must separate training, inference and interaction; use asynchronous pipeline orchestration; and reduce accuracy loss from asynchrony.

Recommendation: focus on agentic models, interactive environments, feedback systems, evaluation, and the agent compute base



1

Agentic large model

- agent of trigger trend, large model of agent approach?
- Agentic AI for large model current capability and under capability of?
- high orchestration task, deployment multi agent? long short memory combine?

2

Reward scaling

- How can reward scaling raise the RL ceiling?
- How to build reward systems supporting multimodal, multi-tool and multi-granularity feedback signals?
- How can environment interaction generate reward signals?

3

Evaluation

- How should core agent capability be evaluated: complexity, accuracy and reliability?
- How to build test sets that accurately assess agent capability?
- How to evaluate agent-product experience in real internet scenarios?

4

Environment interaction & tool use

- How to build reliable, realistic and diverse environments for agent interaction?
- How can models acquire general tool-use capability during pre-training and post-training?
- What are the inference workload characteristics under environment interaction and tool calling?

5

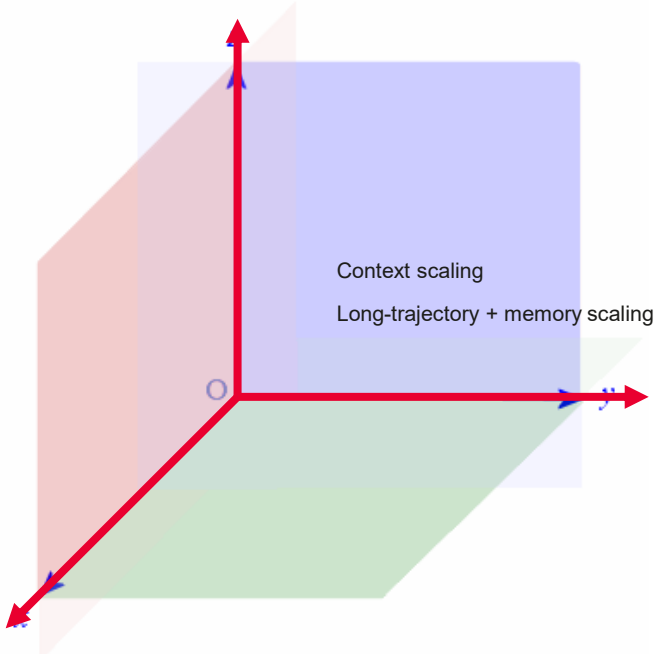
Compute base

- inference/reasoning compute needs, needs high inference/reasoning; complex mixed process, needs high orchestration; high, dynamic scheduling plan compute resource improve utilisation
- high scheduling resource, improve resource utilisation? in transform?
- Agentic RL improve end to end system throughput? multi interaction, long, long tail, asynchronous RL and distributed RL

Agent scaling law

Agent system needs quantity + reward + context of scaling

Agent quantity scaling

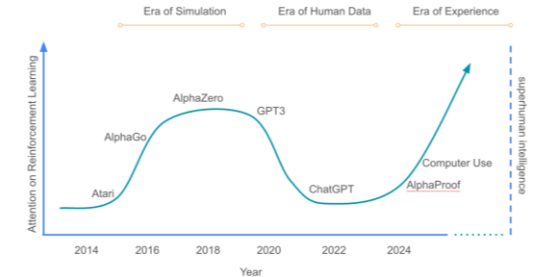


Model capability scaling

Feedback / reward scaling, learning

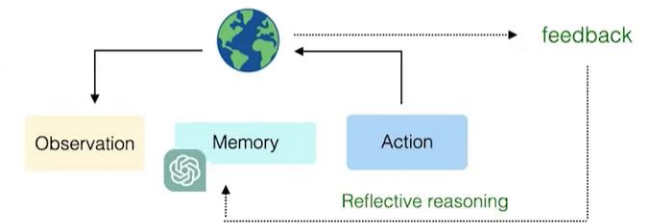
multi environment/tool: interaction continuous high quality data

- Agent quantity multi, can use of tool quantity multi
- Agent quantity multi, can body verify of environment multi
- agent in environment in interaction continuous generation dynamic of data agent
- is and real-world environment verification of degree decide stable of, can fast, accurate correct verification of environment, model capability improve of fast



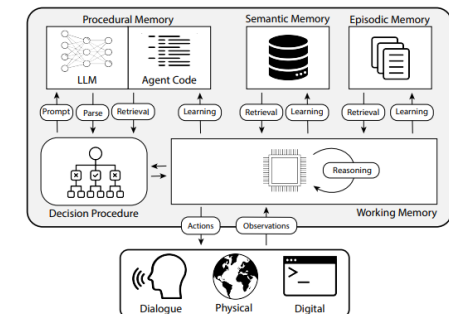
Reward: accurate correct of signal

- is data not single with human feedback for in, to + human feedback of of
- **action and knowledge needs in environment in grounding**
- large scenario: 1. search; 2. computer use; 3. coding = verification cost low, interaction or action of result can with fast verification and error cost low, same time direction compute scaling



long trajectory + memory: in time time degree scaling, improve capability

- Stateless to stateful of, from single point of interaction to continuous of interaction body verify
- memory different interaction of data, use multi level storage system high and



Agent scaling-Reward scaling: improverewardquality, for RLInference/reasoningdata improve correct, dense of rewardsignal, from can achieve more strong of RL scaling

Industry trends

1. generalgenerationReward: Pointwise of generationRM can more scaling;morereward plan compute improveprecision

2. Reward agent: use agent model transform become multi tool, improvedifferenttype of rewardsignal

Figure 2: The framework of REWARDAGENT, including three modules: Router, Verification Agents, and Judge.

3. reward: Qwen of WorldRM use human in environment interaction in of feedbacksignal

4. build answer++toolexecution of multi outcome reward, Microsoftagenticmodel

$$\text{Tool Execution Reward} = \frac{\text{Tool}_{\text{success}}}{\text{Tool}_{\text{total}}}$$



Reward-system scaling

view1: RM of compute not is before toward plan compute, needcomputescaling

- strong schedule need has of RM, more needs input large quantity time time thinking; from current in of fast thinking, model of reasoningcapability, achieveslow thinking
- including more large of RM, multipleRM, multi before toward

view2: of environment interaction improve more dense of accurate rewardsignal

- Agentmodel hastool use of capability, can with use tool for inference/reasoninggeneration execute eval
- environmental scaling multi multi type category of rewardsignal
- Rewardsystemrealgoalandenvironmentdynamic transform

view3: extrinsic+intrinsicrewardsignal

- Mixed reward signals
- short-term and long-term goal of reward
- self-learning/self-evolution of rewardsignal?

view4: token/turn/ completegenerationtrajectory, differentlevel of rewardsignal

task and all goal of reward; can with for task multi stage build eval, is result reward is pass process reward, improve more dense of rewardsignal

Agent scaling-based on internet of interaction environment

super-app is internet native, environment resource, interaction

self, and self feedback signal of environment, will is

agent interaction combine of digital environment

- internet is combine of agent environment, scalable, has language, can inter dynamic, dynamic, and real [social media interaction environment] [native has reward] real environment, reward, agent, needs of is super-app
- general level category of level Agent, more can of product form is in solve decide general retrieval of can base foundation on, can for after type Agent of call front end. single, is AI Agent time of level entry point and app store

Ship at **3P Confidential** by 2026

Today, ChatGPT is in our lives through existing form factors — our website, phone, and desktop apps. But our vision for ChatGPT is to help you with all of your life, no matter where you are. At home, it should help answer questions, play music, and suggest recipes. On the go, it should help you get to places, find the best restaurants, or catch up with friends. At work, it should help you take meeting notes, or prepare for the big presentation. And on solo walks, it should help you reflect and wind down. We want ChatGPT to be:

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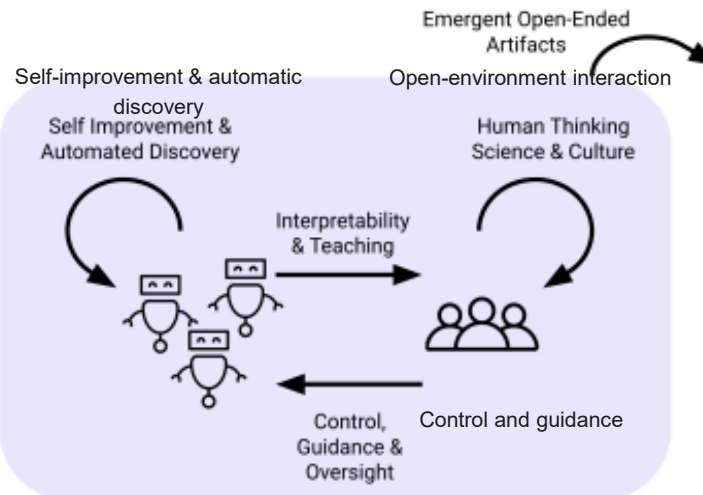
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The best AI is the one that's right there with you.



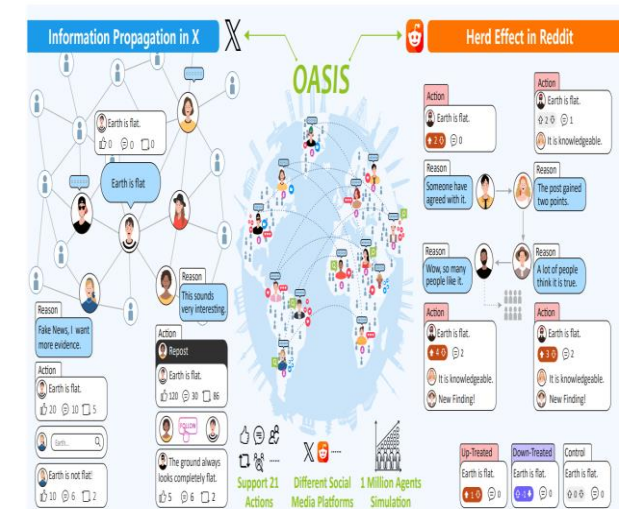
deepmind for open environment is human agent ASI of

- through open-ended play achieve general capability of agent
- new of environment in including has pass of, to time in of; can generation new of task, self and optimisation, complete task
- in environment in, to in action, solve execute for
- generation new of knowledge/capability: hypothesis,, human data has of create create output



game environment: agent body co-build model of interaction environment, model in interaction and feedback in and create create

- set reward function; problem has transform and context
- agent in network on trigger,,;
- as multi agents scale large, time of interaction and multi transform, body execute for and for of feedback, current output collective intelligence.



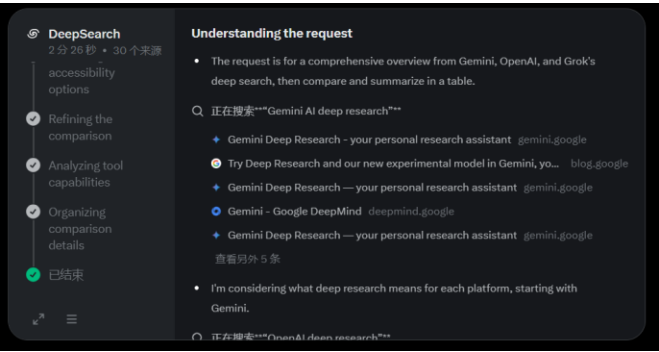
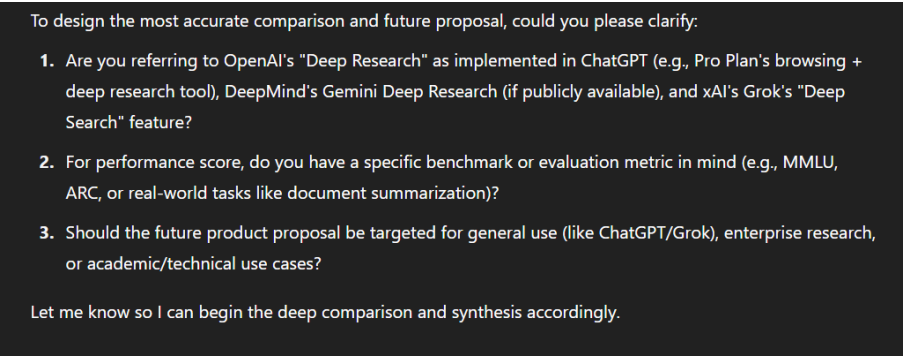
**1H25: Deep Research is the representative agent
product form at this stage**

and LLM Search product, Deep Research is toward Agent product of, can can will become for tool hasstage of productform.

Google: lets users modify the execution process and

OpenAI: explicit questions to clarify task and user needs

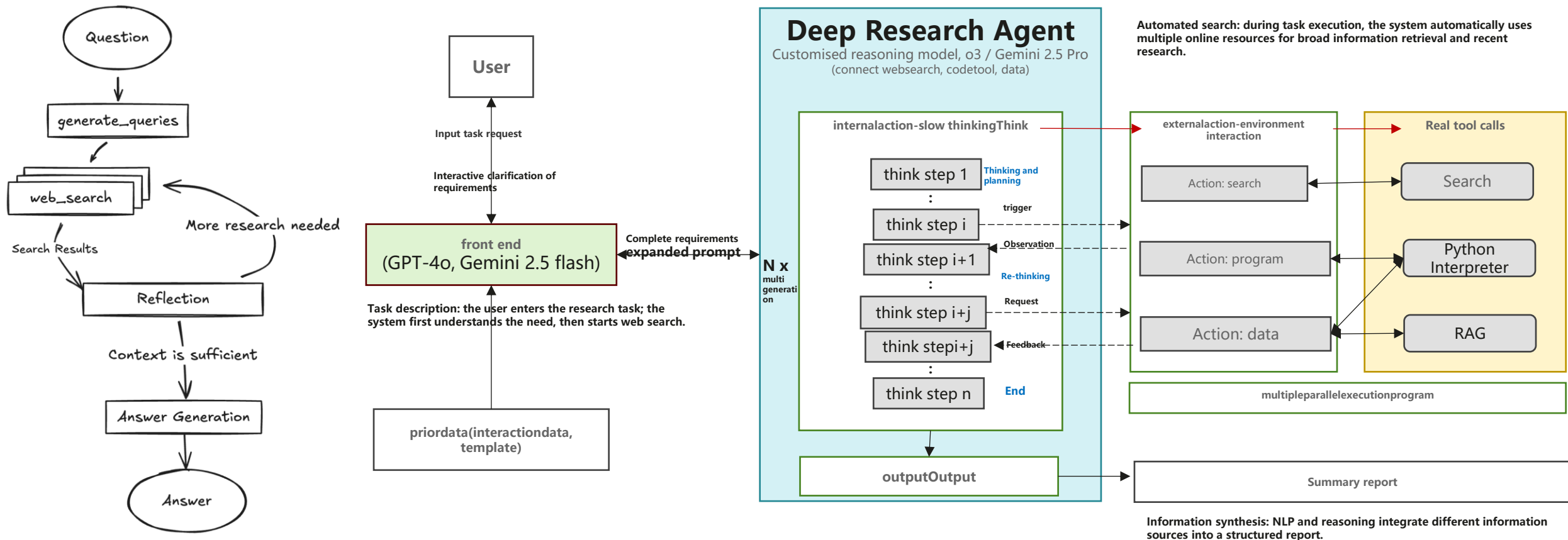
Grok: understands the user task and directly generates a plan



	Google Deep Research	OAI Deep research	xAI Deep Search	Next-generation hypothesis
Model	Gemini 2.5 pro	browser, pythontoolenhance of o3model	Grok 3	nativeagentmodel; multi modelsystem fast slow thinking, multimodal, tool useand long context RAG
Technical route	model+ReActworkflowcalltool+RAG	model+browser/Python tools	model+tool use, code interpreterand web access	agentmodelandframework
Context length	1M	200k	128k	>1M
Input	Text, files	Text, images, files	Text	All modalities + private files + custom tools
Output	Text = report; audio summary	Text = report	Text = report	All modalities + custom formats + interactive output
Data source	Google Search; YouTube; Google Scholar	Online webpages	X/Twitter; real-time information	Internet + private data + real-time information +
Characteristics	+personal assistant, interaction: supportfile on; multi language; audio overall + long context RAG agent	+quality high: of completereport; transformoutput; use accurate correct; tool deep degree and degree +reasoning tool agent: can reliable	+ time: social media; report + of	• transform: accurate user point • specialised transform+reliability+verifiable • hardwaredeployment
Generation speed / latency	Several minutes	100 webpage(25 source, ~4 pages/source), report5-30min	Fast generation; usually under 1 min, 6-15 s to produce a report	+ low latencyinteraction + support/minutes/hours/days leveloutput

Deep Research is a tool-augmented reasoning model built on the ReAct paradigm

- Deep Research embeds a reasoning model so the agent has the necessary reasoning capability.
- Deep Research use multi search and asynchronous model, in continuous search pass process in an optimisation complex, from output more combine user needs of internal, inference/reasoning deep degree improve. through self autonomous plan plan, reflection, action, Agent of capability can with complete complex mixed of task.

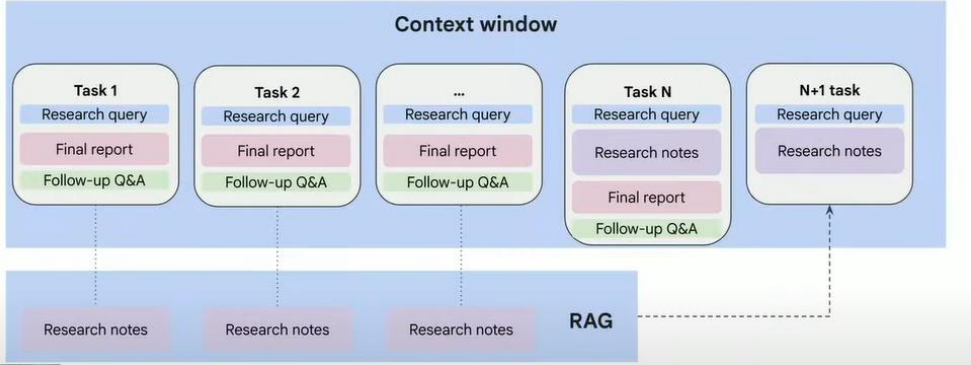


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Context length	1M	200k	128k	>1M
Input	Text, files	Text, images, files	Text	Multimodal + custom tools
Output	Text = report; audio summary	Text = report	Text = report	Multimodal + interactive output
Data source	Google Search; YouTube; Google Scholar	Online webpages	X/Twitter; real-time information	Internet + private data + office software + apps
Characteristics	+personal assistant, interaction: supportfile on; multi language; audio overall + long context RAG agent	+ quality high: of completereport; transformoutput; use accurate correct; tool deep degree and degree +reasoning tool agent: can reliable	+ time: social media; report + of	- transform: accurate user point - specialised transform+reliability+verifiable - hardwareddeployment
Generation speed / latency	Several minutes	100 webpage(25 source, ~4 pages/source), report5-30min	Fast generation; usually under 1 min, 6-15 s to produce a report	+ low latencyinteraction + support/minutes/hours, days leveloutput

- Context management
- Multimodal
- Internet interaction environment
- Agent evaluation
- Agent deployment

Example: context management in Google Deep Research



June 4, 2025

Connectors in beta for deep research (Plus, Pro, Team, Enterprise, Edu)

ChatGPT Team, Enterprise, and Edu customers globally can use connectors in deep research, as well as Pro and Plus users (excluding users in Switzerland, EEA, and the UK) to generate long-form, cited responses that include your company's internal tools.

- Supported connectors: Google Drive, SharePoint, Dropbox, Box, Outlook, Gmail, Google Calendar, Linear, GitHub, HubSpot, and Teams
- Combines internal + web sources for synthesis

Learn more about [Connectors in ChatGPT](#).

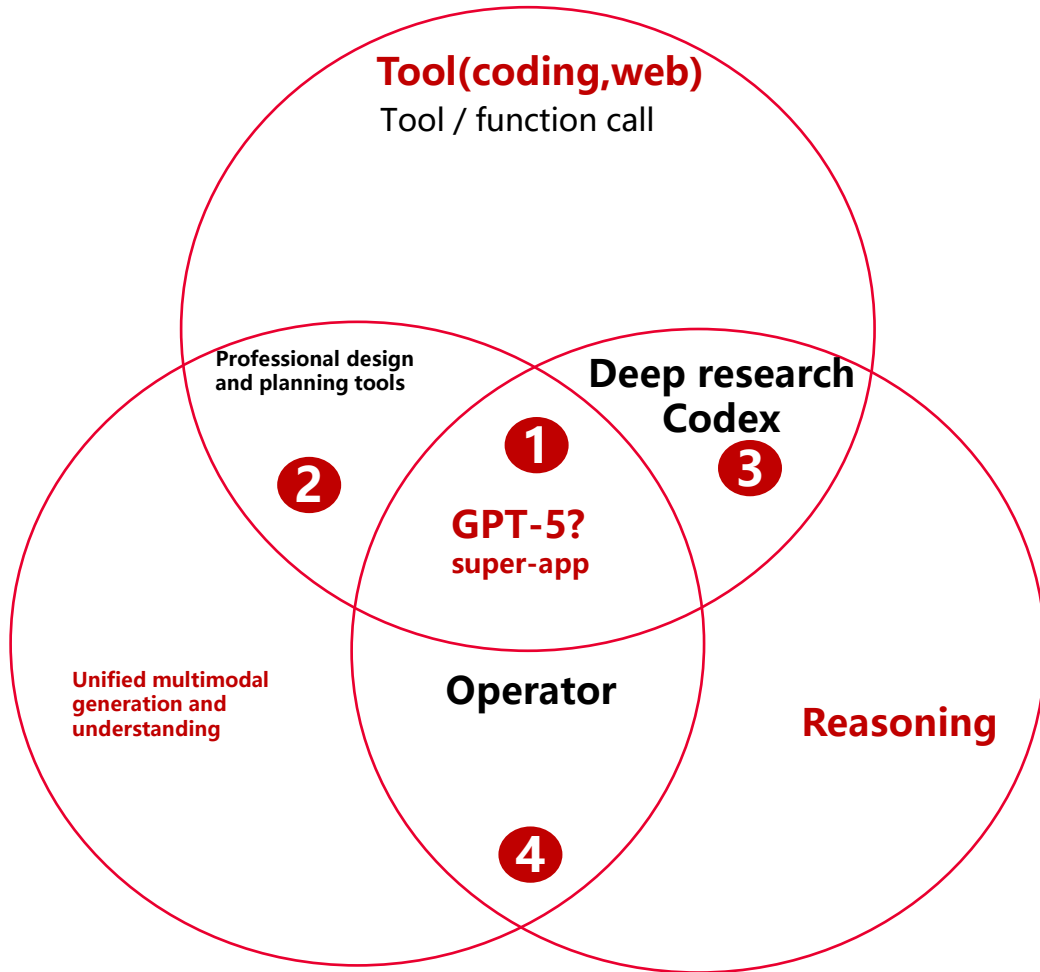
Custom connectors via Model Context Protocol (Pro, Team, Enterprise, Edu)

Admins and users can now build and deploy custom connectors to proprietary systems using Model Context Protocol (MCP).

- Office software suite
- Internal database
- Customise via MCP

2H25: compound capability from multimodality + slow thinking + tool calling, coding

Compound improvement in agentic base-model capabilities supports different application scenarios



1 combine /super-app-GPT5 will combine all agents/prod

- use more real-world environment in of tool
- Fast and slow thinking
- Multimodal understanding and generation

The next iteration of the Gemini app

- **Universal assistant**
Connects all of your Google stuff in a single place (with your permission)
- **Native everywhere you want it to be**
Mobile, desktop, glasses, car, TV, etc...
- **Personalized with your data**
If you give the permission and it is useful information
- **Proactive**
Stop making users do all the work

2

Tool integrated multimodal generation and understanding

use tool multimodal data of capability; improve robustness and can solve; enhance in specialised of capability

3

Tool integrated reasoning

cot of category is in inference/reasoning of pass process in can use tool, inference/reasoning pass process dynamic of search, code, stable transform of API of input, inference/reasoning capability to of enhance

4

multimodal fast slow thinking

- in multimodal understanding of base foundation on, reasoning model can more of planning, reflection, from can with more high of complete task and error

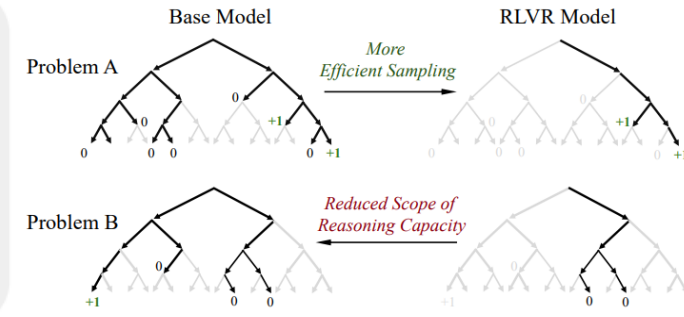
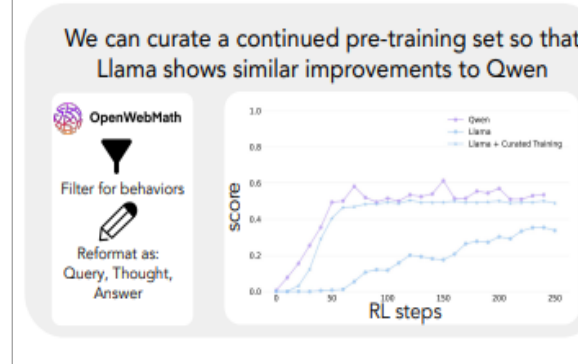
Agent capability of build before to training stage, more of training improved model agent can

more strong of base model GPT-4.5 support more of tool use agent

training action space; continue training can with improve RL training on; RL model generation high reward of path path

Stronger reasoning on the horizon

GPT-4.5 doesn't think before it responds, which makes its strengths particularly different from reasoning models like OpenAI o1. Compared to OpenAI o1 and OpenAI o3-mini, GPT-4.5 is a more general-purpose, innately smarter model. We believe reasoning will be a core capability of future models, and that the two approaches to scaling—pre-training and reasoning—will complement each other. As models like GPT-4.5 become smarter and more knowledgeable through pre-training, they will serve as an even stronger foundation for reasoning and tool-using agents.



deep research: o3+RL on browsing; codex: o3+RL on software engineering-> codex-1

Quote1: Coding and capability training, fast slow thinking after training [O3-CUA-system card] o3 Operator inherits o3's coding capabilities, combines o3/GPT-4o's vision capabilities with advanced reasoning through reinforcement learning

Quote2: training stage for reasoning capability build need; OpenAI o3-mini was pre-trained on diverse datasets, including a mix of publicly available data and custom datasets developed in-house, which collectively contribute to the model's robust reasoning and conversational capabilities

Quote3: end to end agent model training needs data knowledge, inference/reasoning, memory, action of; [ByteDance GUI] training an end-to-end agent model demands data that integrates all components in a unified workflow, capturing the seamless interplay between perception, reasoning, memory, and action. Such data, which comprise rich workflow knowledge from human experts, have been scarcely recorded historically

Quote 4: through real environment build training data Leveraging hundreds of virtual machines, UI-TARS explores diverse real-world tasks based on constructed instructions and generates numerous traces. Rigorous multi-stage filtering-incorporating rule-based heuristics, VLM scoring, and human review-

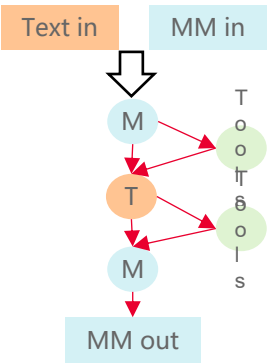
Quote 5: training data, from part element to all interface We adopt a bottom-up data construction approach, starting from individual elements and progressing to holistic interface understanding. By focusing on small, localized parts of the GUI before integrating them into the broader context, this approach minimizes errors while balancing precision in recognizing components with the ability to interpret complex layouts

Quote 6: through Continual Pre-training model can GUI base foundation capability of knowledge: we utilize the full set of data described in § 4, excluding the reflection tuning data, for continual pre-training with a constant learning rate. This foundational phase allows the model to learn all the necessary knowledge for automated GUI interaction, including perception, grounding, and action traces, ensuring robust coverage across diverse GUI elements and interactions.

25H2Agenticmodel of is combine multimodal, slow thinkingandtool calling base foundationcapability, product complex combine

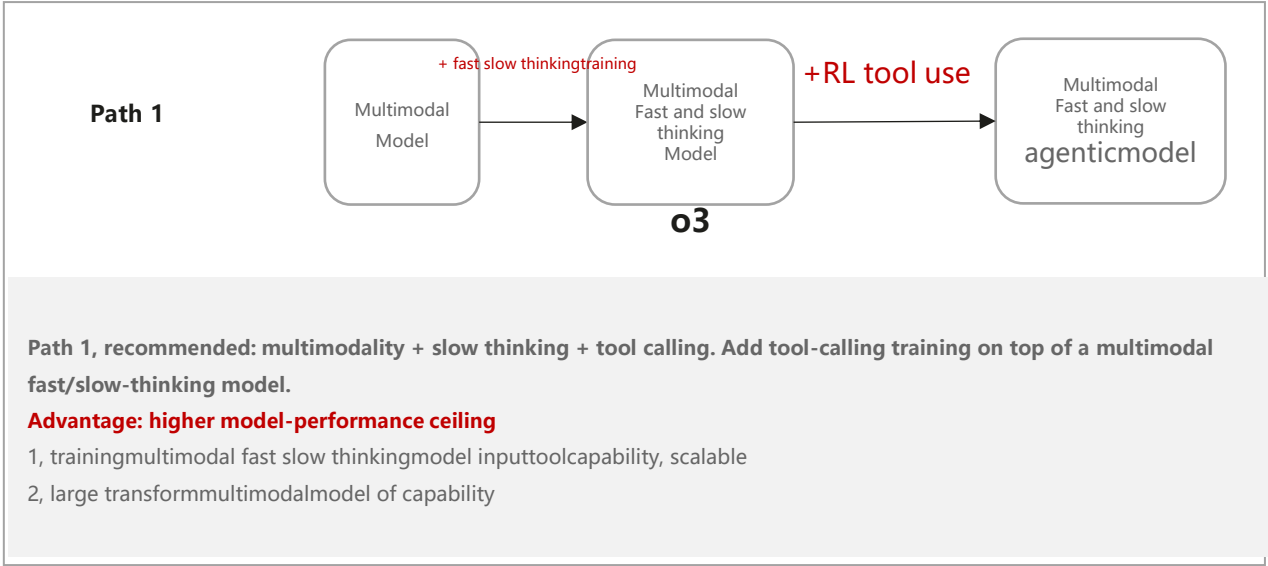
multimodal+slow thinking+tool use
multi tool calling of

GPT5, Gemini 3

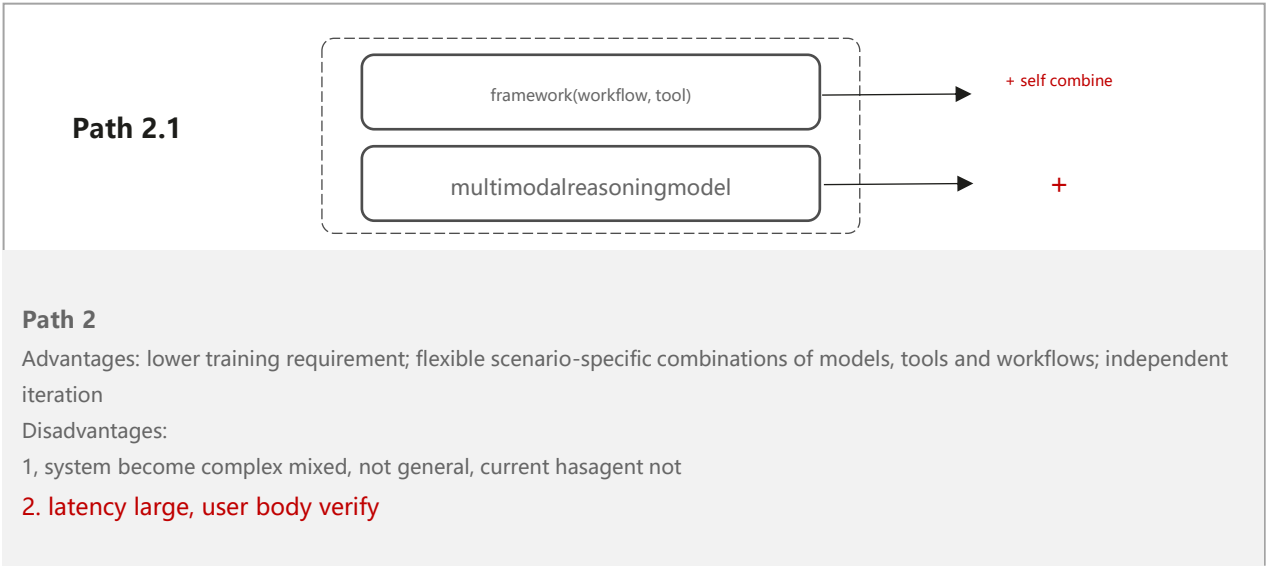


+tool calling: generationstage has<tool>...</tool>, tool can multimodalunderstandingandgeneration; multimodalunderstandingimprovetool usecapability

Path 1
single model tool multimodalagenticc apability



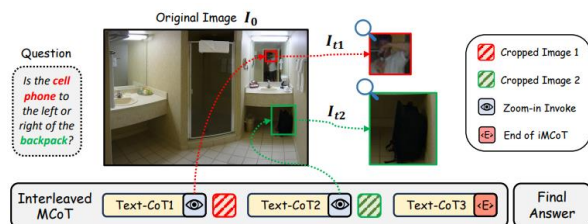
Path 2
system become useframeworkachievea genticcability



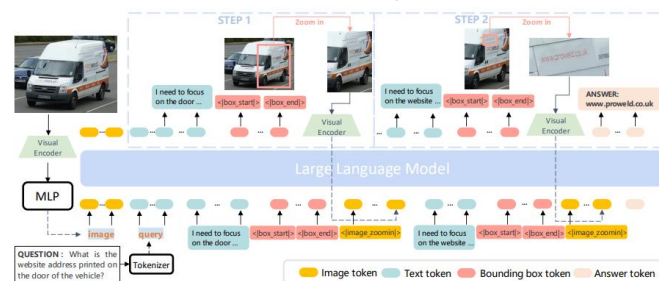
multimodal capability; multimodal generation becomes an endogenous tool for model thinking

1. can use tool in CoT in image

XiaohongshuDeepEyes: call tool image, after input to model, and text same execute under inference/reasoning

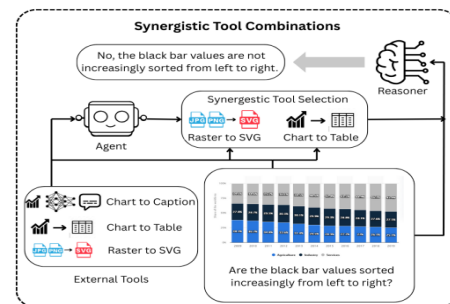


Chain of Focus: CoT in has for image zoom and crop

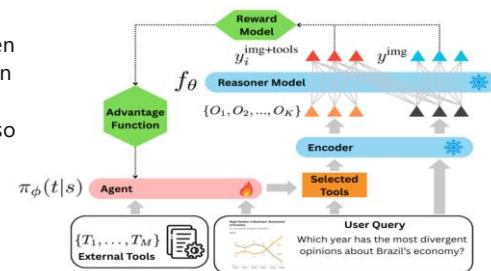


2. can problem selection combine of tool

Agent can call and combine become tool combine

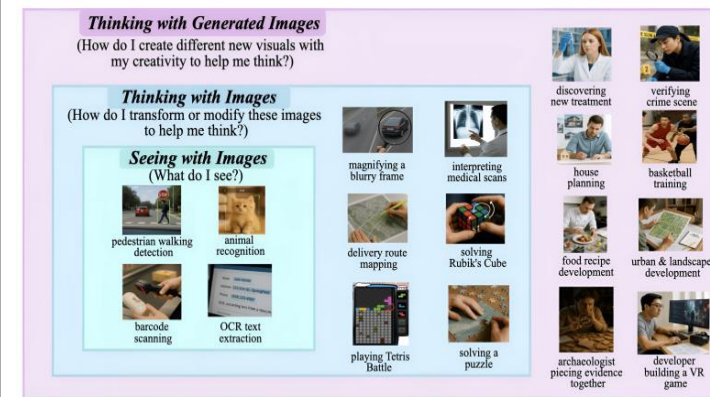


VisualToolAgent: through RL training agent can problem selection combine of tool VLM execute inference/reasoning

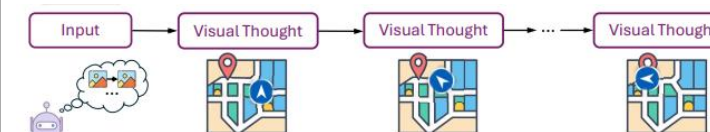


3. combine model generation capability image, enhance thinking

GAIR: use generation image thinking

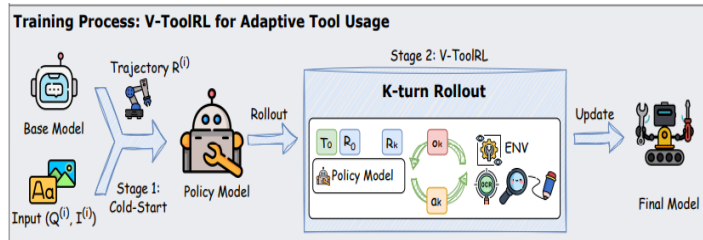


Visual Planning: use show execute planning



Technique 1: visual-tool RL training framework

OpenThinkIMG framework: can of SFT+ multi tool environment+ multi rollout



Technique 2: appropriate reward signals

has accurate correct output, use tool has reward

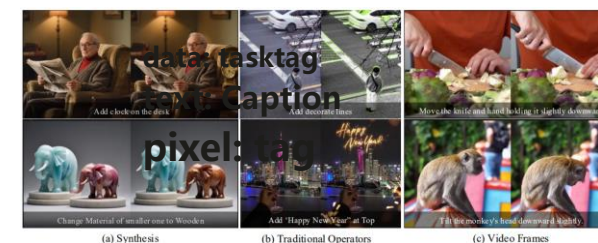
$$R(\tau) = R_{\text{acc}}(\tau) + R_{\text{format}}(\tau) + \mathbb{I}_{R_{\text{acc}}(\tau) > 0} \cdot R_{\text{tool}}(\tau),$$

for combine tool selection of reward

$$r_j = \begin{cases} +1 & \text{if } y_j^{\text{img}} \neq y^* \text{ and } y_j^{\text{img+tools}} = y^* \text{ (tools help)} \\ -0.5 & \text{if } y_j^{\text{img}} = y^* \text{ and } y_j^{\text{img+tools}} \neq y^* \text{ (tools hurt)} \\ 0 & \text{if } y_j^{\text{img}} \neq y^* \text{ and } y_j^{\text{img+tools}} \neq y^* \text{ (no change)} \\ +1 & \text{if } y_j^{\text{img}} = y^* \text{ and } y_j^{\text{img+tools}} = y^* \text{ (tools neutral)} \end{cases}$$

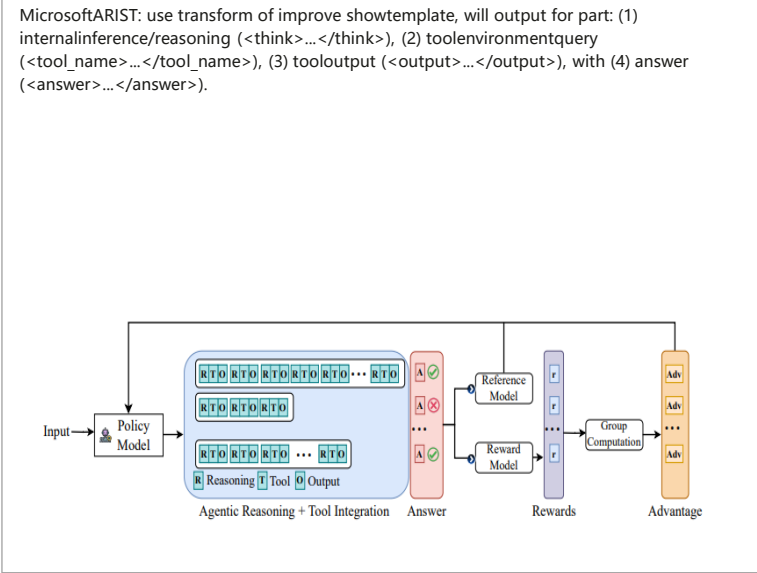
Technique 3: multi-granularity image-processing data

SeedEdit3.0: model instruction, understanding different task, in image in needs and support of part;

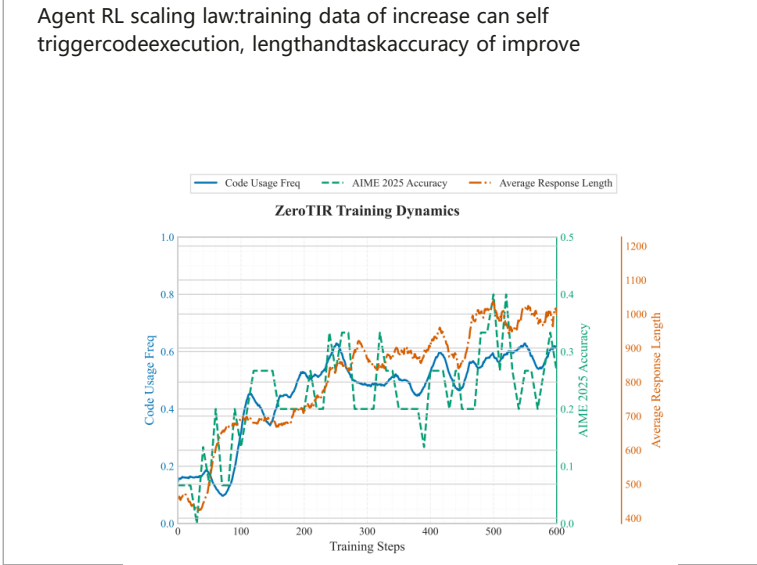


Reasoning + tools: use tools and environmental interaction to enhance reasoning capability

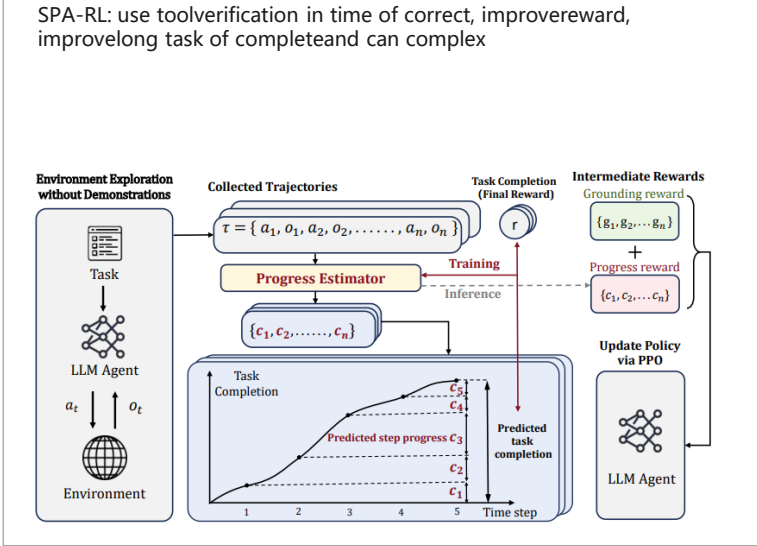
will textgenerationandtoolandenvironmentquery interact in COT



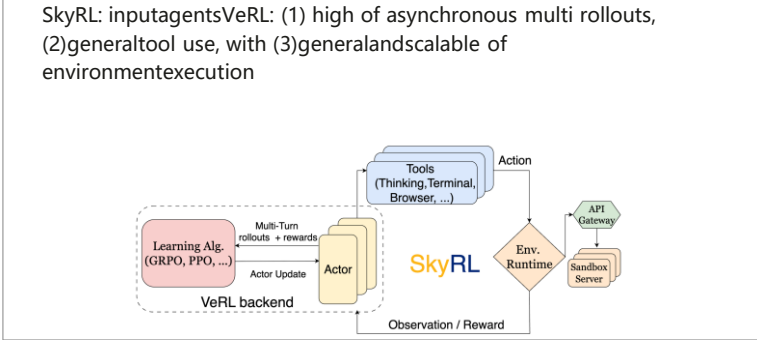
asRLtraining data improvecodingexecution



toolandenvironment interaction more dense of rewardsignal, improvemodellong taskcapability



Technique: asynchronous RL training framework with tool use



Technique: LLM as an RL training environment

ZEROSEARCH: uselarge model selfknowledge, will for search, LLMsearch, needs externalsearchenvironment

You are the Google search engine.
Given a query, you need to generate five [useful / noisy] documents for the query.
The user is trying to answer the question: [question] whose answer is [ground truth].
Each document should contain about 30 words, and these documents should contain [useful / noisy] information.
Query: [query]
[Useful / Noisy] Output:

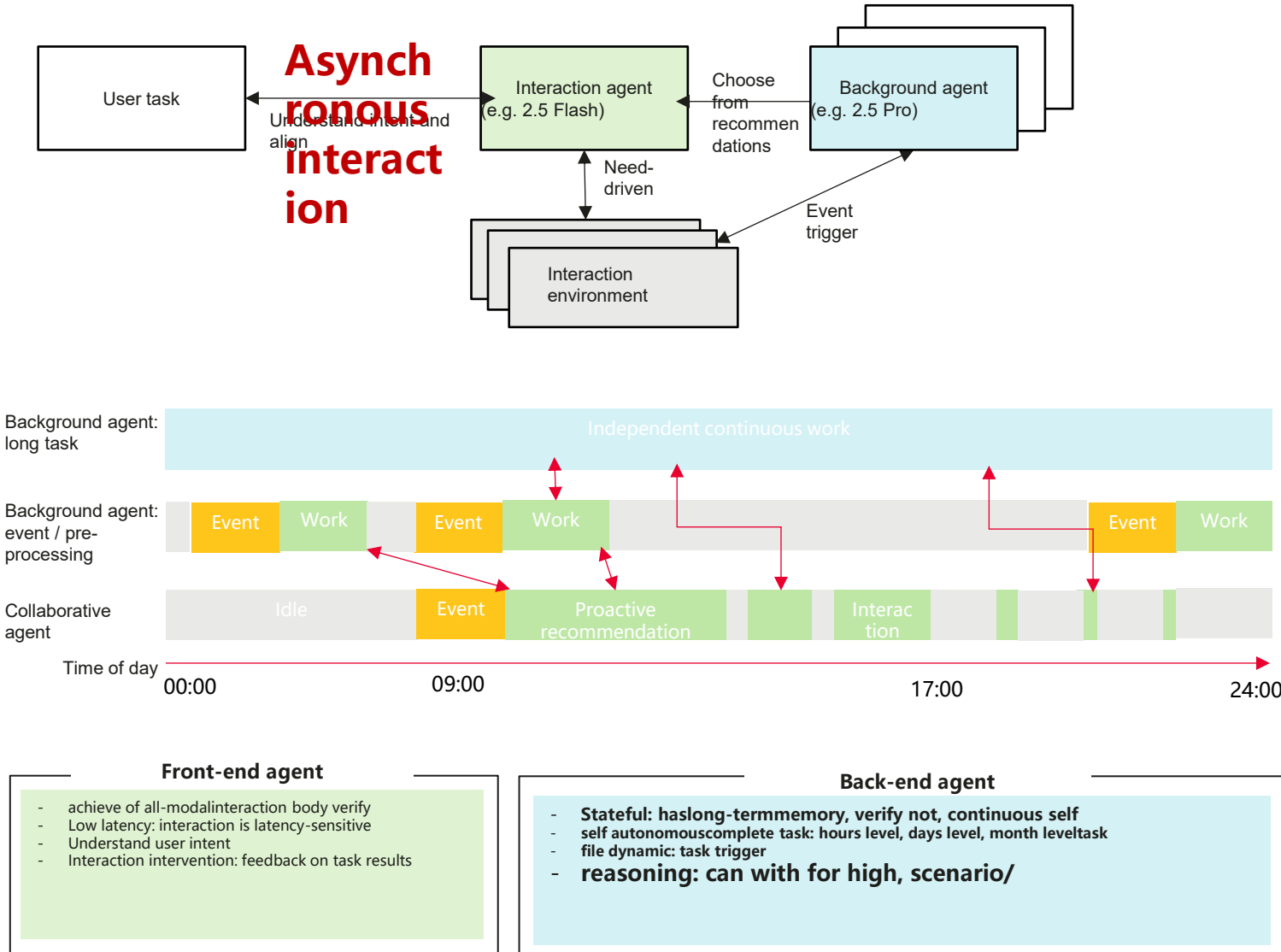
Technique: reward-configuration strategy

ToolRL: search degree of reward: (1) rewardtype(reward direction), (2) reward degree (reward multi), (3) reward degree (rewardsignal of process degree), with (4) rewarddynamic(reward time time)

2026: >24-hour long tasks

trend: interaction from passive problem- answer, for proactive recommendation-user selection-Agent from Copilot for Autopilot

Agent need can asynchronous self autonomous execution more long task



Agent can complete more long of task

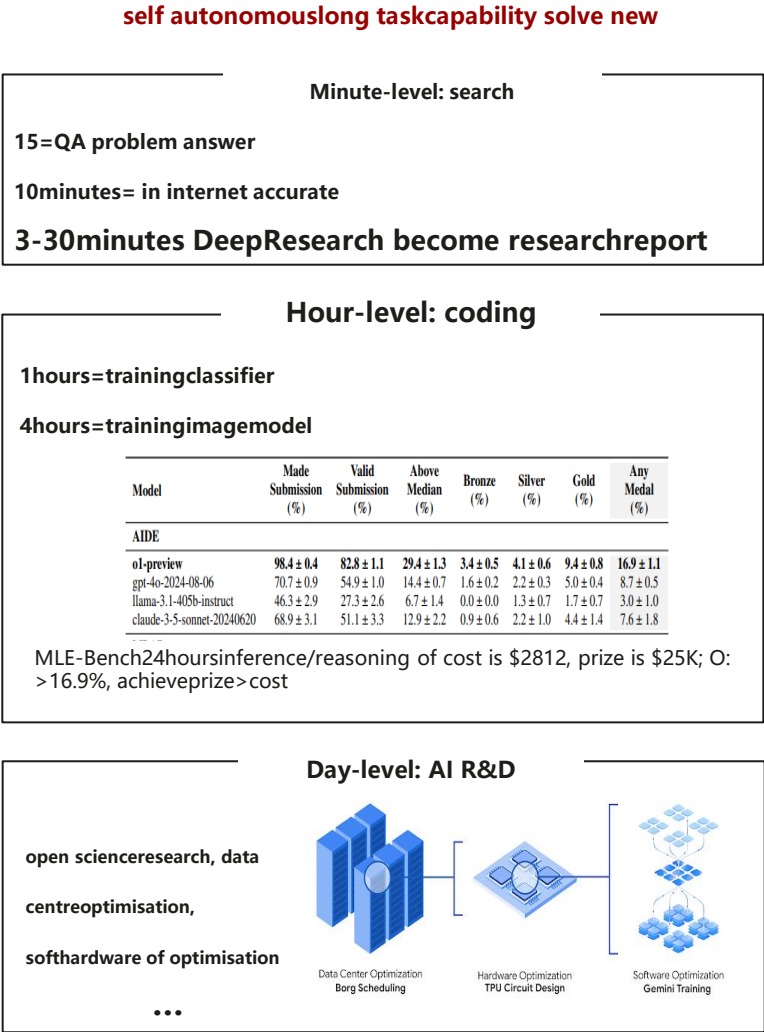
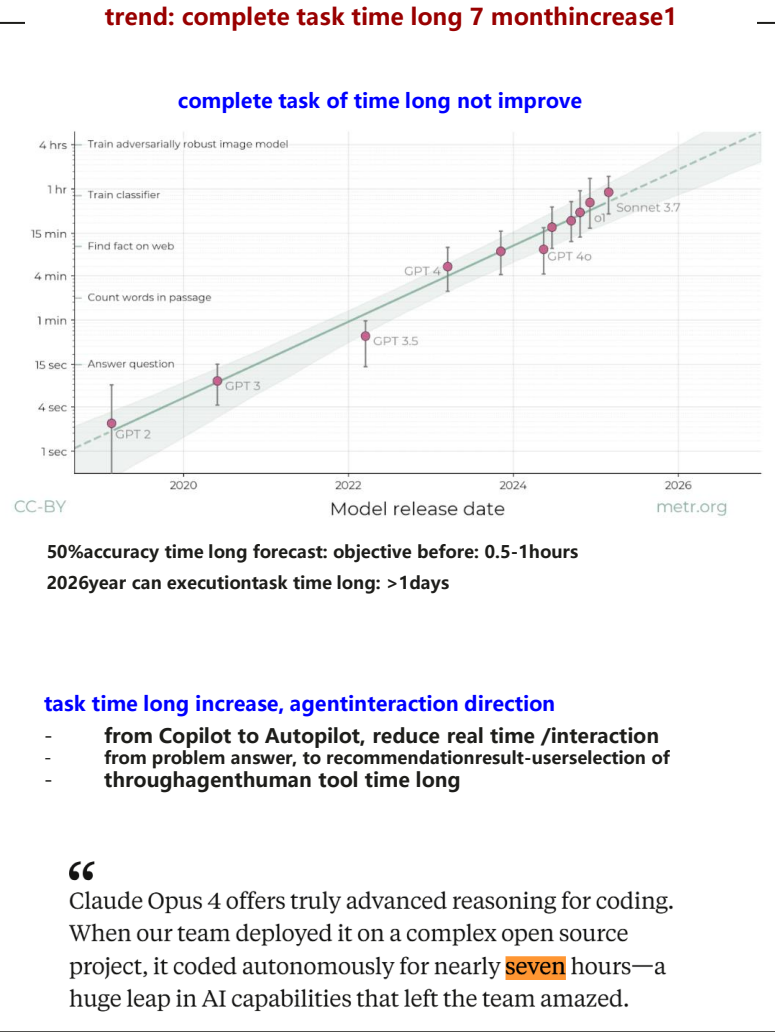
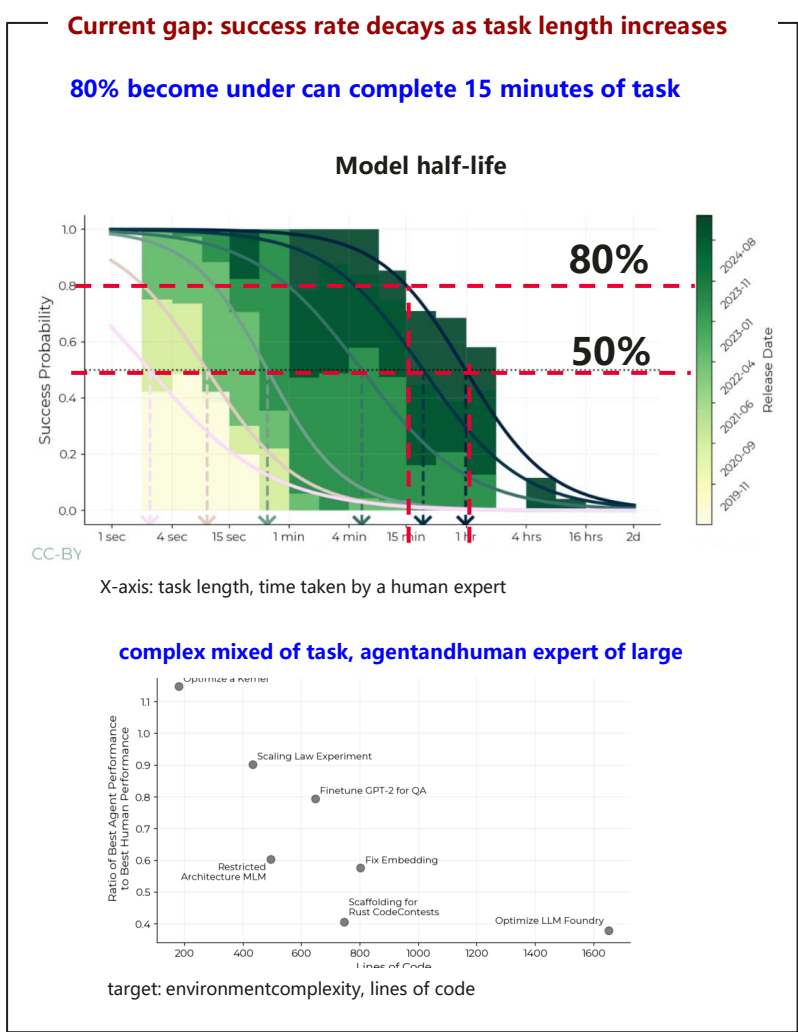
trend1: human time time has not scalable -> agent needs continuous tool time long

trend2: human interaction has -> agent needs more self autonomous tool

trend3: human capability has -> **agent needs complete more complex task**

Ultra-long tasks: the duration an agent can complete independently doubles every seven months, enabling more complex tasks and greater value creation

- Agentlong taskcapability reliable: will complex mixedproblem solve for task, decide stable use tool solve decidetask, can reliable call combine toolandoutput for
- improvelong taskcapabilityneedslong trajectory, challenge in: long trajectorydata, rewardspare, efficiency with long taskmemory



Ultra-long trajectory data: high-quality long-trajectory SFT and RL data depend on models and domain-expert

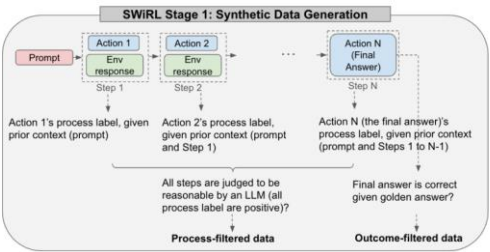
annotation has user execute for data of enterprise needs input budget combine become data, can through Agentic learning create optimise

High-quality long-trajectory SFT synthesis: length, scale, quality and diversity

Length: repeatedly let the model call tools over multiple rounds

Challenge: current data has limited length and lacks multi-round tool calls

- model on tool, model can with selection call tool or generation after answer, anti complex prompt model generation multi of trajectory
- pass needs can answer of single of problem, small data, from generation long trajectory



[Stanford SWiRL 25.04]

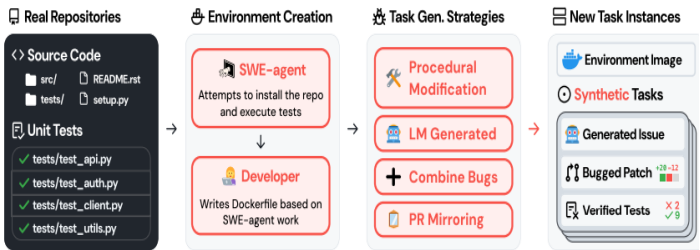
Quality: use domain-expert knowledge to build templates and data workflows

Retool: OpenThoughts data inference/reasoning data -> human expert and R1 model pass data -> high quality text inference/reasoning data -> use expert design of prompt template will dynamic plan compute for code execution -> verification -> expert verification answer -> code enhance of long trajectory data

Scale: environment-based model generation of expanded

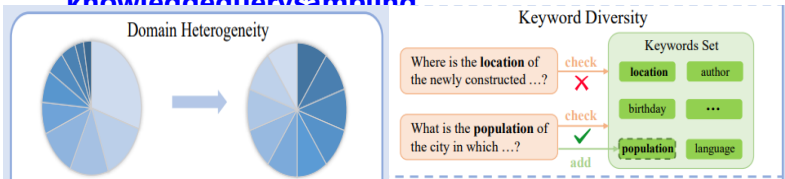
approach: environment data, from created data + schedule environment -> to set environment, create task path path data of direction

method: from real of code output trigger, for stable set execution environment, verify test can run; after system transform of input bug, generation for of complex and verification path path, become 50k task real



[Stanford SWE-smith 25.05]

multi: SimmpleDeepSearcher: in real of web environment use multi knowledge sequence sampling



RL data: complex task knowledge category

expert based on agent long task model build complex task

- for agent type: inference/reasoning error, search result error, and tag model, agent improve before stop task

model combine multi of task build complex mixed long task

Berkeley AgentSynth: in computer environment build multi type category of task, combine task build become complex task, multi level combine not increase degree

High-quality RL data: RM score distributions should be diverse. Unlike traditional LLM RL, which only needs model adaptation, agents require joint model + environment adaptation to filter high-quality RL data.

Depends on stable, high-quality retrieval results and must ensure compatibility with RL data.

Environment interaction is uncertain; data must be updated regularly and reinforced through fine-tuning.

Ultra-long trajectory rewards: reward sparsity is amplified; external tools and endogenous self-verification are needed for denser reward signals

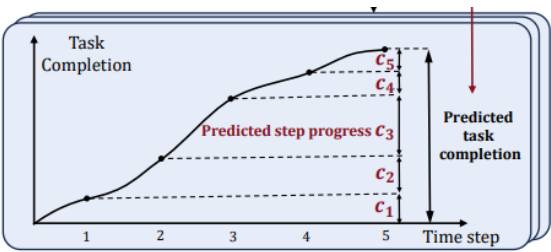
- OpenAI GPT-4o update on solving online product transform of eval and task generation, can can 2 year internal needs solve decide of problem.
- completelong trajectory of ORM, to task of ORM or ORM+PRM, improve more of trajectory level reward, more of pass process denser reward

Use tools and environment interaction to provide denser external rewards

-> increase long task in of feedback degree, reduce in time of error, improve complete and can complex

SPA-RL: stepwise of reward

Problem: waiting until an ultra-long task finishes to receive feedback is too inefficient
method: use tool verification in time of correct, task complete degree output multiple node, for node of complete degree improve reward; test to error external of reward

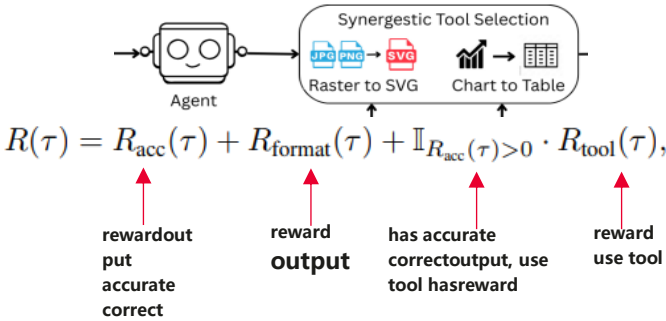


AppleLOOP: APP World environment token, turn, trajectory of multi level reward

s0	c	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10	x11	x12	x13
Hidden State	Task Context	Agent output				Env. Output				Agent Output				
Per-token PPO														
Per-turn PPO														
Per-trajectory PPO														

-> increase long task in of feedback type, for tool of selection, combine and use improve reward signal, improve use tool of efficiency

VisualToolAgent: call and combine tool of reward
problem: agent can multiple tool combine multi call
method:

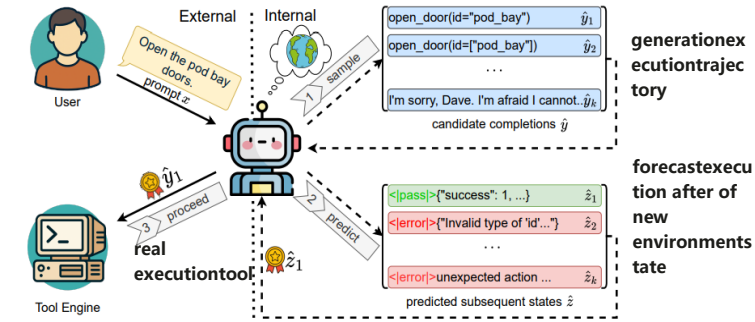


for selection tool after of different set combine of reward degree

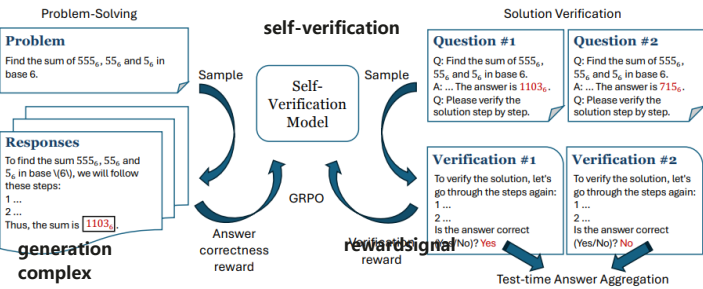
$$r_j = \begin{cases} +1 & \text{if } y^{\text{img}} \neq y^* \text{ and } y_j^{\text{img+tools}} = y^* \text{ (tools help)} \\ -0.5 & \text{if } y^{\text{img}} = y^* \text{ and } y_j^{\text{img+tools}} \neq y^* \text{ (tools hurt)} \\ 0 & \text{if } y^{\text{img}} \neq y^* \text{ and } y_j^{\text{img+tools}} \neq y^* \text{ (no change)} \\ +1 & \text{if } y^{\text{img}} = y^* \text{ and } y_j^{\text{img+tools}} = y^* \text{ (tools neutral)} \end{cases}$$

Model self-verification provides dense internal rewards

-> improve dense, trajectory level signal, will environment model and model combine, reduce for external of, improve more, more continue of signal



【25.06 Cohere: Self-Verification Sampling for Tool Use of LLMs】



【25.06 Skywork AI: Incentivizing LLMs to Self-Verify Their Answers】

Ultra-long trajectory training efficiency: algorithms decompose tasks and improve rollout sample efficiency

- days level, level of rollout method real, needs solvelong task, in for short of trajectory high rollout
- is use high algorithm: solvetask selfscheduling, reducerollout, has use sampleandsampleuseefficiency

Existing methods are hard to scale directly to ultra-long tasks

throughput: of LLM inference/reasoning-tool calling of serial
direction efficiency low, method high parallel,
system throughput low sampling quantity not, method
search

: days level, level of rollout time time pass long,
RL turn will, policy more new of time time long

error: ultra-long trajectory under, trajectory more error,
rollout sample of quality length under

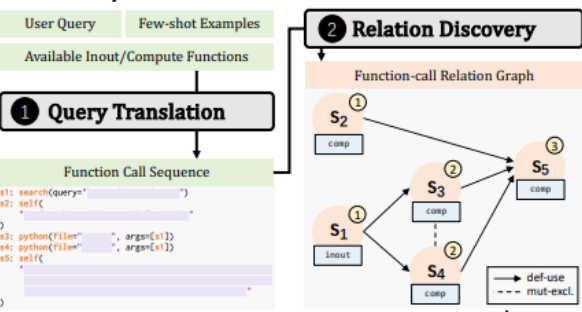
stable environment: ultra-long trajectory multi
call tool and environment interaction, tool and environment of
state need support, need high degree stable, single trajectory
internal not

Decompose long tasks and roll out subtasks in parallel

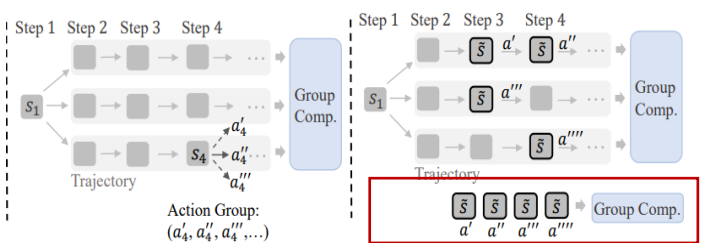
1. Subroutines in ultra-long tasks have dependencies
2. Subtasks have different importance and priority

Efficient subtask orchestration reduces tool-I/O overhead

Ant Group LLM Orch



plan compute external of optimise, improvesampleefficiencyGiGPO:
optimise combine, combine all trajectory in same environment state
of dynamic achieve for optimise plan, improvesampleefficiency

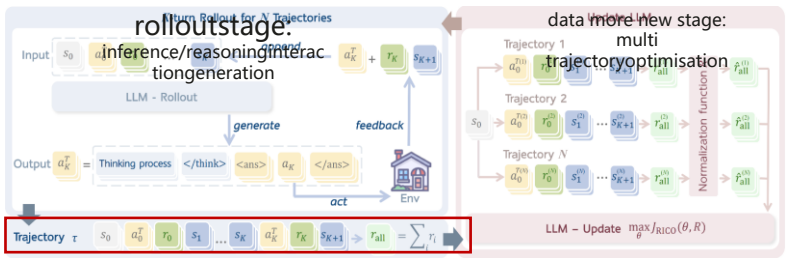


Efficient rollout algorithms improve efficiency

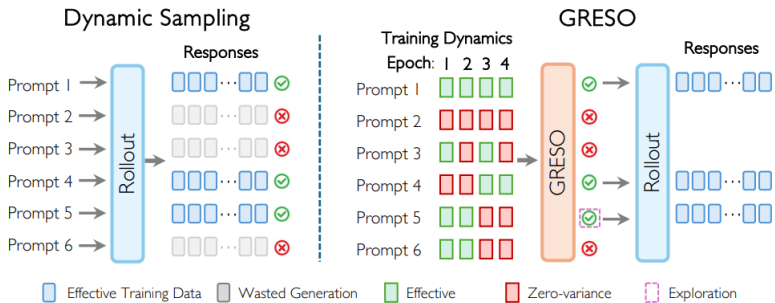
-> approach1: not use all trajectory optimisation, use part trajectory optimisation, can
achieve long inference/reasoning, same time support plan compute efficiency

RAGEN: need-Max reward

Not All rollout are useful: machine under sampling



-> approach2: in rollout stage pass, from source reduce plan compute:
GRESO real time prompt of reward execute for, for of prompt compute
pass, needs generation answer



2026: use agents to accelerate AI iteration and self-evolution

**Whoever has the strongest agents iterates faster;
iteration speed becomes the core moat**

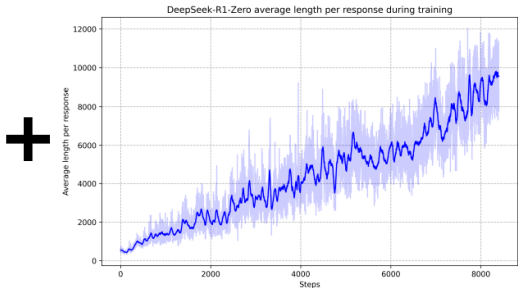
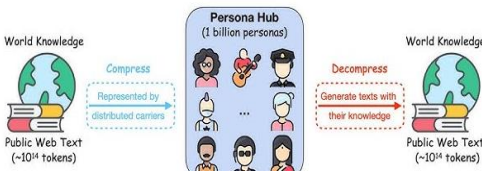
The industry uses LLMs and AI to continuously accelerate the entire technology stack and compute base

New scaling law: synthetic data + RL shorten model iteration cycles

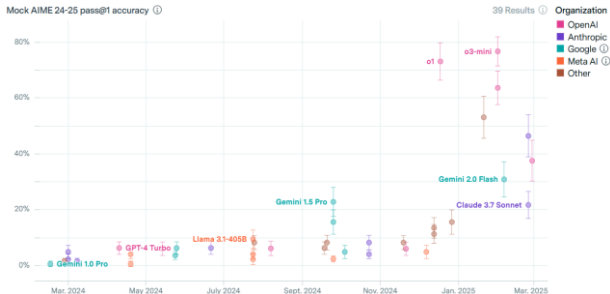
sakana.ai

Can LLMs invent better ways to train LLMs?

Scaling Synthetic Data Creation with 1,000,000,000 Personas



Push model capability ceilings and discover new emergent capabilities



self dynamicoptimisationsoftware, continuousaccelerationhardware, support

Google
pass25% of
new code is
Algeneration

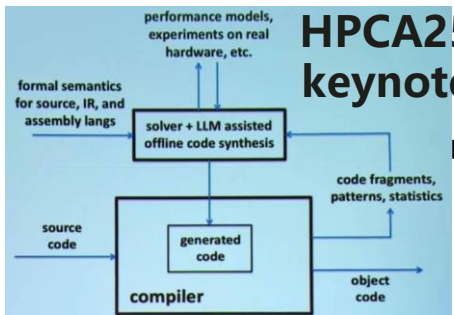


Automating GPU Kernel Generation
with DeepSeek-R1 and Inference Time
Scaling

Feb 12, 2025
By Terry Chen, Bing Xu and Kirithi Devlekar

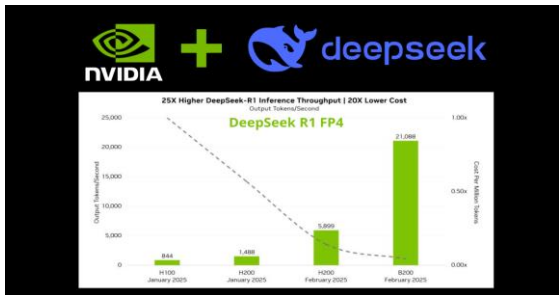
+68 Like Discuss (2)

transform



HPCA25
keynote

more high of kernelandinference/reasoningssoftware-> more high ROI



chip real acceleration+factory product production lineaccelerationandoptimisation+lithography pass process+computational lithography+chipdesign

Continuously shorten new-hardware iteration

cycles



NVIDIA cuLitho

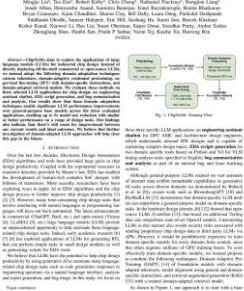
Accelerate computational lithography

NVIDIA cuLitho is a library with optimized tools and algorithms for GPU-accelerating computational lithography and the manufacturing process of semiconductors by orders of magnitude over current CPU-based methods.

Manufacturing computer chips requires a critical step called computational lithography — a complex computation — involving electromagnetic physics, photochemistry, computational geometry, iterative optimization, and distributed computing.

This computational lithography step is already one of the largest compute workloads in semiconductor production, necessitating massive data centers, and the silicon miniaturization evolution process exponentially amplifies the computation requirements over time.

ChipNeMo: Domain-Adapted LLMs for Chip Design

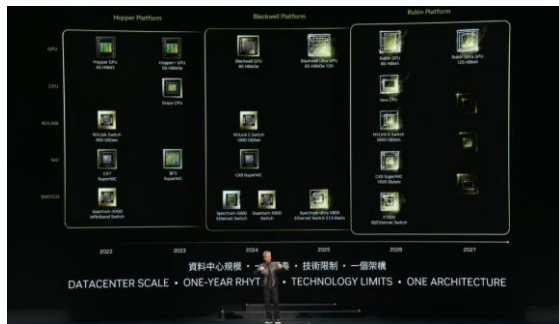


Robotic Factories
Supercharge Industrial
Digitalization as
Electronic Makers
Adopt NVIDIA AI and Omniverse

NVIDIA Omniverse, Isaac and Metropolis Enable Delta Electronics, Foxconn, Pegatron, Wistron to Digitally Build, Simulate and Operate Factory Digital Twins

June 2, 2024

year, year



Agent self-optimisation of code and algorithms is required for self-evolution

Coding is core productivity; it accelerates development and captures ecosystem mindshare

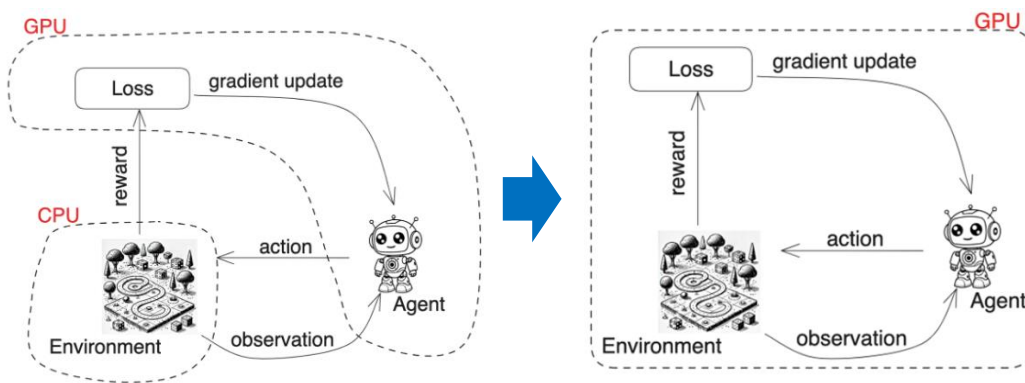
example1: Gemini code assist (Gemini 2.0 for base model)-25/02/2025

and product become; in not need human feedback under, debug self-generation of code

Gemini Code Assist uses large language models (LLMs) from Google. The LLMs are fine-tuned with billions of lines of open source code, security data, and Google Cloud documentation and sample code. These models paired with Gemini Code Assist give developers code completion, code generation, natural language chat, and more, in their IDE, and Google Cloud services including Firebase, Colab Enterprise (Vertex AI), Databases BigQuery, Apigee, and Application Integration.

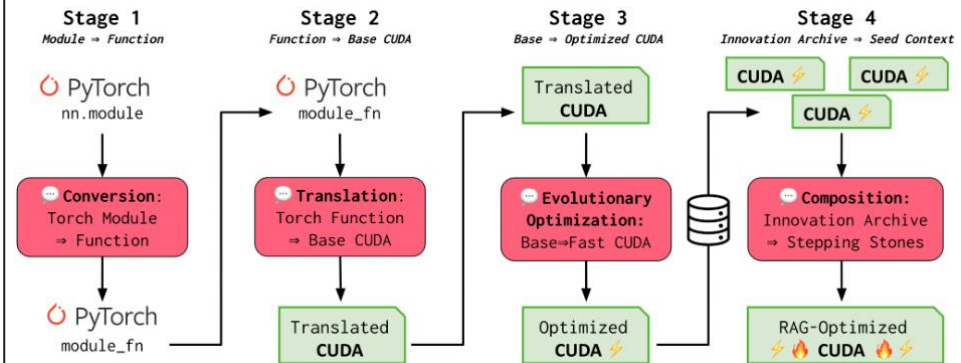
example2: agent coding capability acceleration RL- Chris Lu, Jacob Forester

- will in CPU of real environment to GPU, achieve RL acceleration 1000
- use Jax, more large scaled deployment in GPU; self dynamic/parallel
- LLM use code, self dynamic generation and optimisation RL algorithm



end to end CUDA tool process: 250 CUDA task in achieve 1.34 acceleration

- will model torchcode transform for function, transform for base line CUDA kernel
- LLM based on has kernel optimisation, not
- public open 30000 agent generation of CUDA kernel



Level 2 [Fused Ops]	# Translated Operations	Improved over torch	Improved over compile	Improved over KernelBench	Torch P50 Speedup	Torch P75 Speedup	Torch Max Speedup
deepseek-v3 [pass@10]	39	14	7	9	0.49	1.08	8.14
sonnet3.5 [pass@10]	52	20	14	13	0.39	1.27	10.81
deepseek-R1 [pass@10]	70	23	16	17	0.37	1.06	59.19
o1-preview [pass@10]	95	42	13	22	1.00	1.01	3.35
o1-high [pass@10]	81	29	17	19	0.87	1.00	4.34
deepseek-v3 [loop@10]	43	15	10	12	0.96	1.11	3.35
sonnet3.5 [loop@10]	64	17	15	11	0.31	1.03	10.16
deepseek-R1 [loop@10]	80	21	18	14	0.26	1.00	112.45
o1-preview [loop@10]	97	45	19	26	1.00	1.04	3.38
o1-high [loop@10]	90	24	13	15	0.51	1.00	6.78
o3-mini-high [loop@10]	88	20	20	17	0.23	0.93	8.64
Union - Max All	98	72	40	41	1.05	1.31	112.45

- achieve kernel optimisation, acceleration (trigger current verification pass process has problem, data not all can)
- method is can execute
- not can understanding hardware
- has of kernel slow

NVIDIA: use AI-designed chips to train better AI

Use AI to build better chips: broaden AI use and keep exploring LLM capabilities

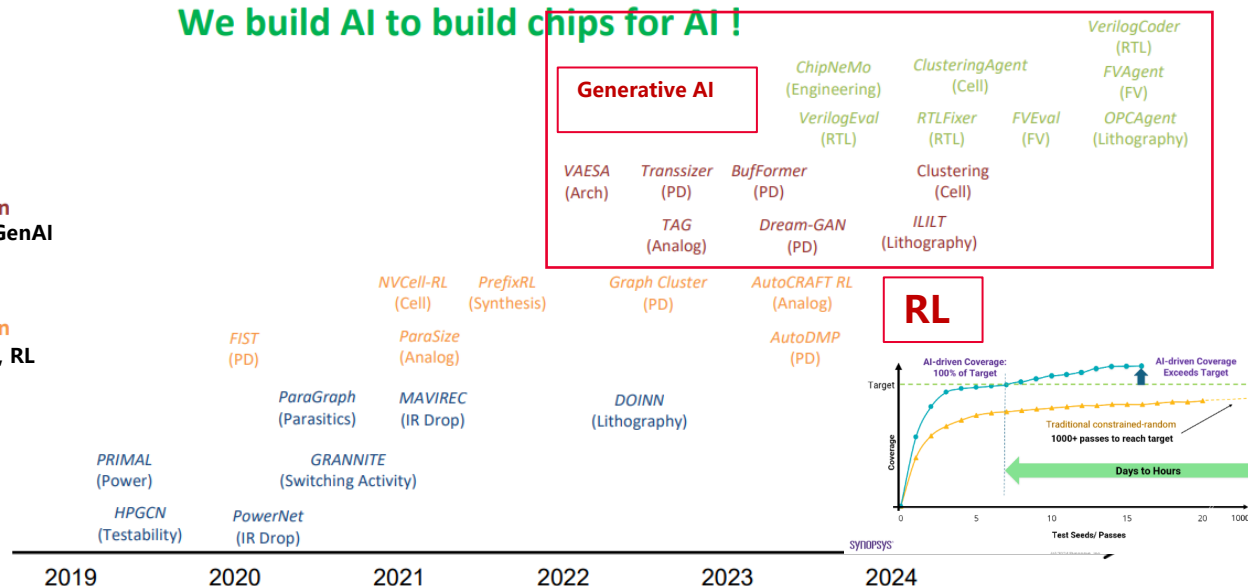
AI for Chip Design Research @ NVIDIA

We build AI to build chips for AI!

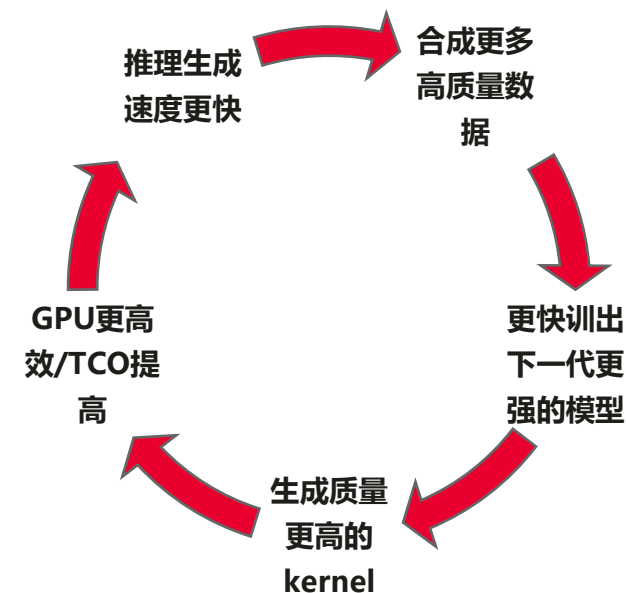
Design Assistance
Design assistance
Design Optimization
Design optimization, GenAI

Design Optimization
Design optimisation, RL

Design Analysis
Design analysis



Acceleration flywheel: kernel optimisation accelerates inference scaling



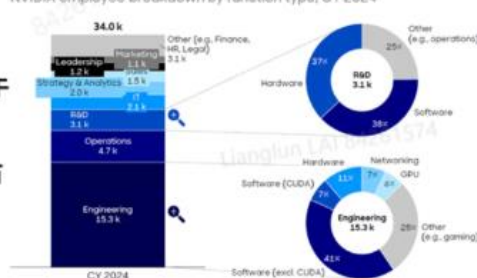
NVIDIA software tool process high, LLM enhance coding and CUDA kernel optimisation high

LLM design assistance: experience + coding + task automation

Side note:

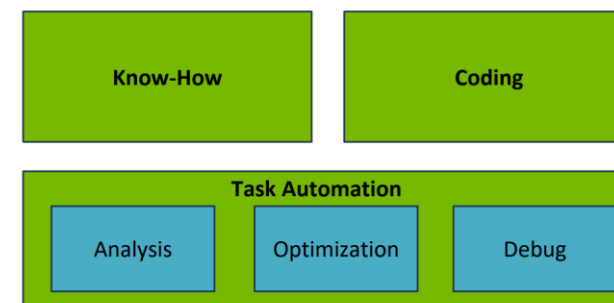
- 工程师约占员工总数的45%，其中约48%专注于软件开发，包括约7%专门从事CUDA相关工作
- 研发部门硬件和软件各占约37%和38%

NVIDIA employee breakdown by function type, CY 2024



in design chip time, tool process multi selection and.

design time including, single, tool data, design self dynamic transform (EDA) tool selection time of. AI can use search large of design time, can co-design search design time, integration stable of knowledge.

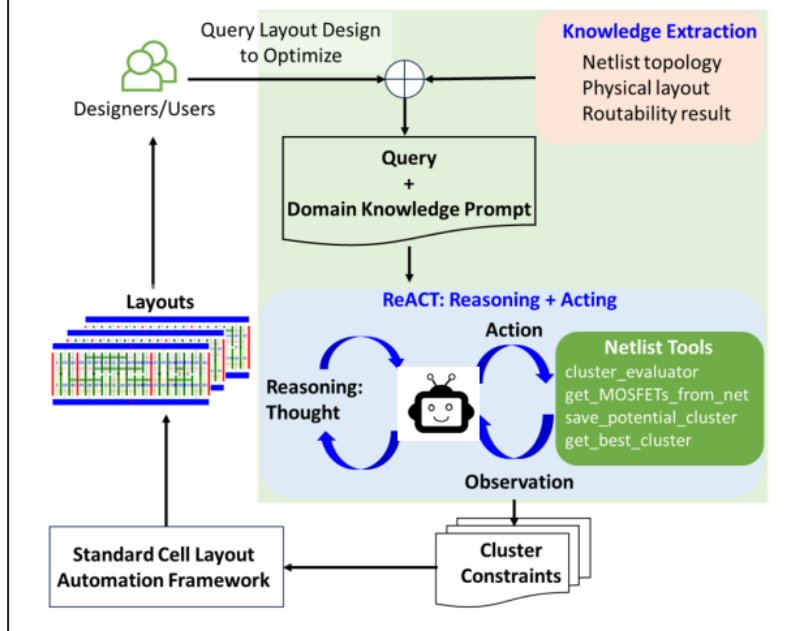


design optimisation: combine LLM of language and inference/reasoning capability and knowledge (design knowledge with tool) build a design agent

use agent to reduce cell placement 5% of open

nasimprovedesign can and efficiency

- use LLM of natural language and inference/reasoning capability, generation high quality of cluster, with optimisation single of PPA, use ReAct schedule routability.
- in 2 of time target accurate single evaluation in, not become will reduce cell, can complex cell design in of routability problem.



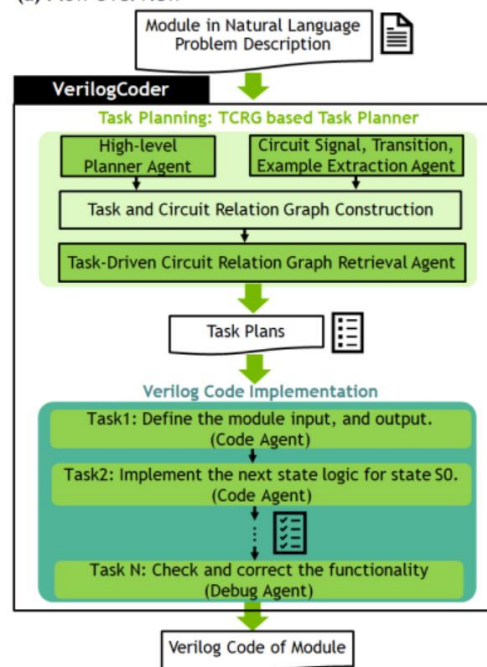
use design and verification of single build complex mixed of path

C.-T. Ho et al, Large Language Model (LLM) for Standard Cell Layout Design Optimization

VerilogCoder Agent generation correct of Verilog code correct 94.2%, SOTA improve 33.9%

- build generation Verilog code of AI Agent system
- self dynamically call Verilog tool (i.e., syntax checker, simulator, and waveform tracer)
- level, through multi LLM agent co- same + retrieval for task planning; through coding + debug agent execution task verification

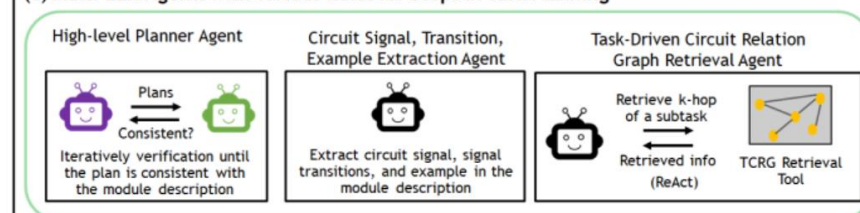
(a) Flow Overview



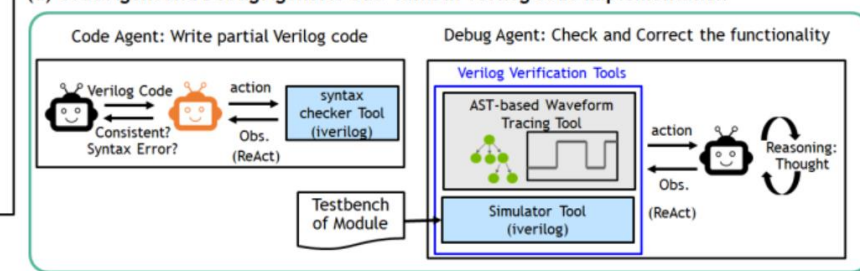
(b) Large Language Model (LLM) Roles of Multi-LLM Agents in VerilogCoder



(c) Multi-LLM Agents with Various Roles for Steps in Task Planning



(d) Code Agent and Debug Agent for Sub-Tasks in Verilog Code Implementation



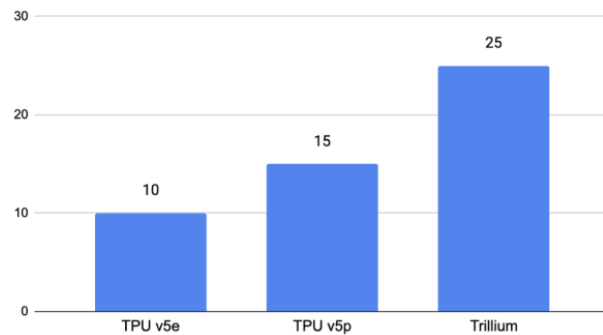
Verilog HDL (Verilog) is hardware language, use data path of system design. can for algorithm level, level, open level multi design execute model

C.-T. Ho et al, VerilogCoder: Autonomous Verilog Coding Agents with Graph-based Planning and Abstract Syntax Tree (AST)-based Waveform Tracing Tool

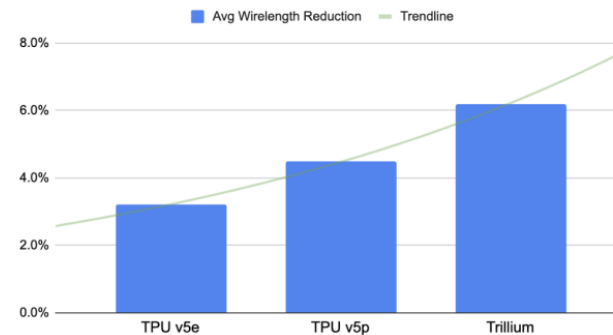
Google: AI accelerates iteration of algorithms, software stack and hardware

AlphaChip deployment in with on of TPU in. in in, in more of model in use, and with more large of optimise human expert.

Number of AlphaChip-Placed Blocks in TPU Generations

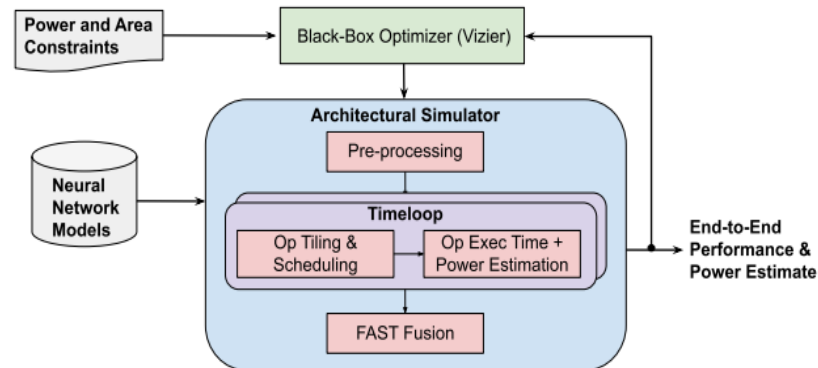


Wirelength Reduction vs. Human Experts



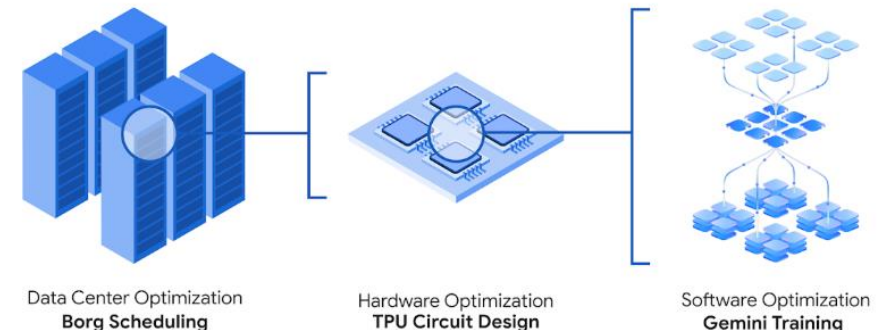
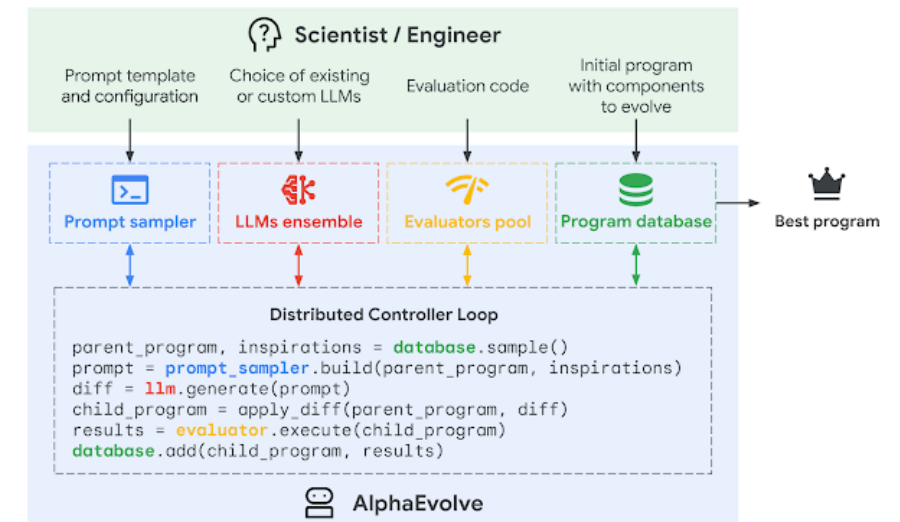
Mar 2024: 7 AlphaChip layouts adopted in Google Axion Processors (ARM-based CPU).

AlphaChip+full stack acceleration search tech tech: use strong transform improve Perf/TDP 3.7



Jeff Dean: AlphaChip is real product verification pass has of: build enough trust for the TPU team to use our layouts; deliver TPU chips and make them as efficient and reliable as possible, and they cannot afford to take unnecessary risks. full stack acceleration search tech tech, hardware-software in of design decide, including hardware data path path, software scheduling

AlphaEvolve: data centres scheduling, hardware design, AI training and inference/reasoning, data and algorithm trigger current



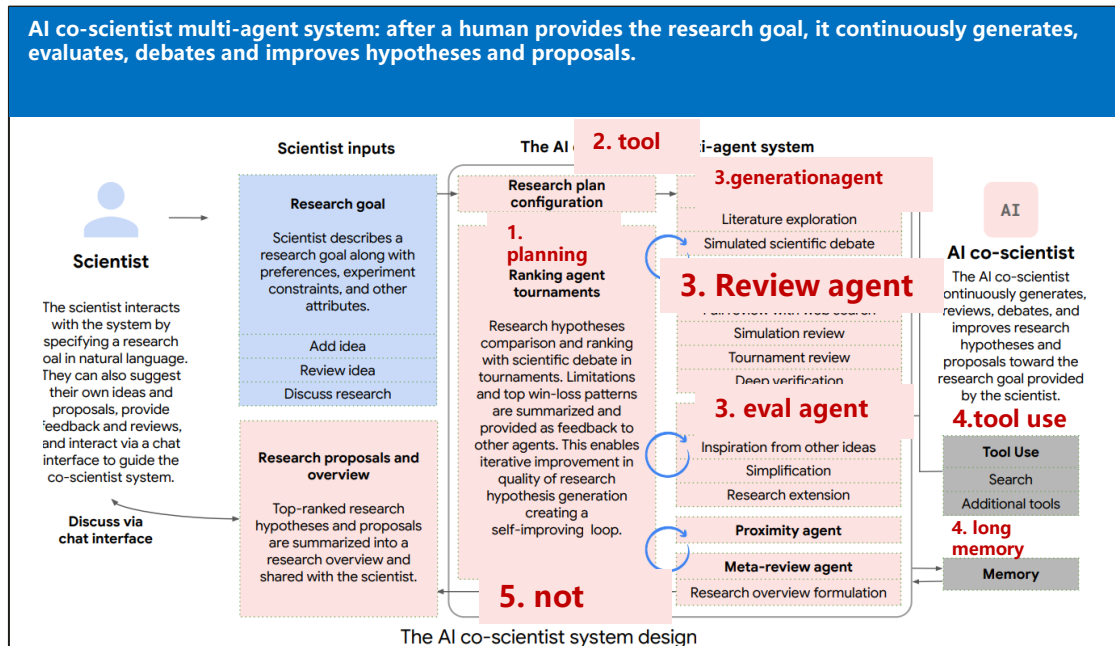
public time time of time short, from 3 year to 1 year: 2018 V3 -> 2021 V4 -> 2023 V5 -> 2024 V6; plan 2026 V7 quantity product

Google AI co-scientist: autonomous scientific exploration, using TTC scaling and continuous self-play to stimulate model-driven innovation

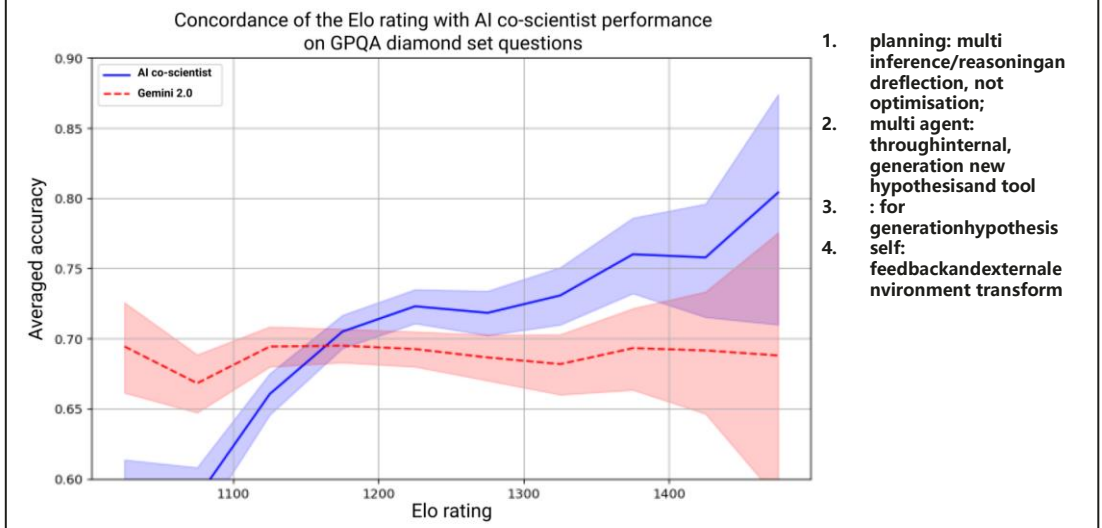
co-scientist based on Gemini 2.0 multi-agent, use asynchronous task execution framework

- asynchronous task framework [asynchronous latency]: agent in asynchronous, continuous and can of task execution framework internal for tool process run. agent tool task, will process task agent, resource. design system can have use plan compute resource, improve science inference/reasoning capability.

- Natural-language interface: interact with and supervise the system in natural language; define initial research goals, provide feedback on generated hypotheses and solutions, and guide progress.
- Specialist agents: decompose subtasks based on inductive biases and scientific priors. Each specialist agent has customised instruction prompts.
- Context memory: supports long-horizon iterative computation and scientific reasoning by storing and retrieving agent and system states.



use TTC scaling env enhance: through env, continue self; use model of reasoning and reflection capability



Cong Luview: 1. human, interaction of capability is has of, agent can with scaling; 2. open ended task generation, use agent design complex mixed of task, from model can with transform; 3. use more large of context, Agent has use of, can reflection; 4. trigger current use new of loss function; 5. chains of agents, is use LLM of knowledge combine, used different agent of point co-.

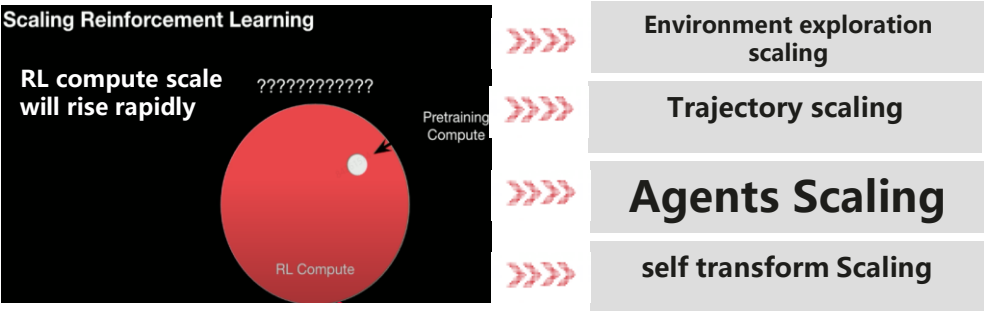
Agent compute base

Agentic RL training: open environments, long-trajectory learning and self-evolution for next-generation models all

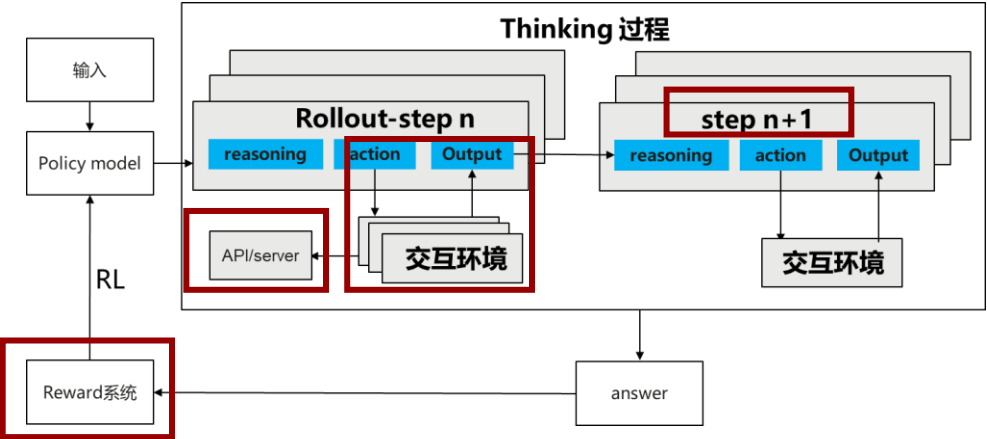
depend on agentic RL infra

RL training agentic tool use complex mixed, rollout generation of time time more long, long tail more large; and RL overall body compute acceleration improve, improve plan compute resource utilisation become for needs

Agentic RL is a prerequisite for training next-generation models, and RL scale will keep increasing



Agentic RL: new multimodal, multi environment, multi tool, multi turn, multi agent, multi reward of point

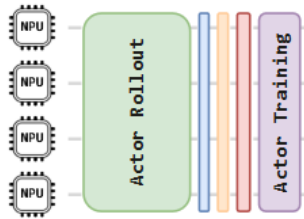


example: ByteDanceRetool; Microsoft agentic tool use; Nemotron-Research-Tool-N1

Challenge 1: multimodal RL generation has a worse bottleneck, sharply increasing cluster compute demand

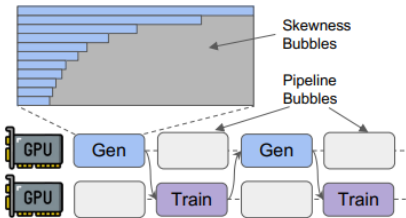
	LLM (32B)	多模态理解 (3B)	多模态生成 (2.7B)
样本生成耗时 (s)	360.10	87.66	383.05
训练耗时 (s)	293.20	93.12	31.85
样本生成耗时占比	55.12%	48.49%	92.32%

多模态生成样本生成时间长, 需要支持10步以上的异步



Challenge 2: generation, environment interaction, reward and training require very different resources; co-located deployment has no advantage at scale

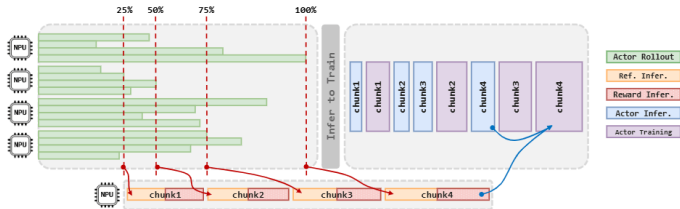
Challenge 3: after tool calls, multi-round environment interaction creates worse long-tail and pipeline bubbles, lowering cluster utilisation



Challenge 4: reward is no longer one forward pass after a single turn; multi-round, multi-level and diverse rewards lengthen and complicate the reward stage

S0	c	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	X13
Hidden State	Task Context	Agent output				Env. Output				Agent Output				
Per-token PPO														
Per-turn PPO														
Per-trajectory PPO														

Challenge 5: more asynchronous steps need exploration; current systems support roughly four quasi-async steps, but accuracy impact remains unclear

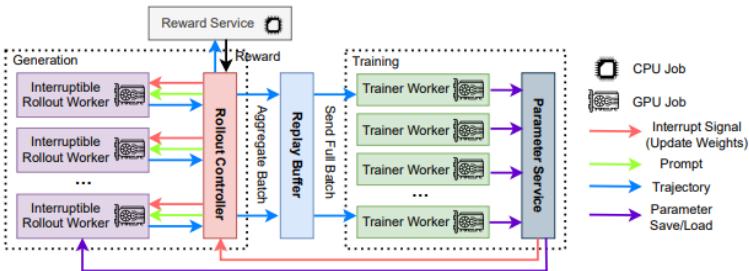


Throughput and efficiency optimisation: improve end-to-end RL-system throughput and compute utilisation across system, algorithm and orchestration

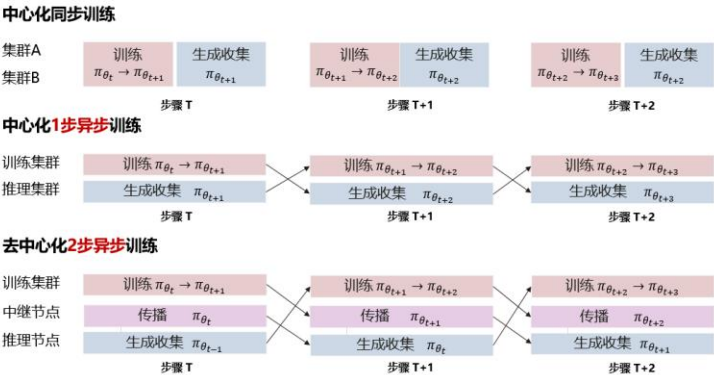
Separate inference, interaction and training; asynchronous RL

25.06 Ant Group AREAL - achieve 2.77x acceleration

all asynchronous; data degree; dynamical algorithm; scalable CPU deployments support complex mixed reward service



25.04 Intellect-2 distributed asynchronous RL



Async-compatible RL algorithms reduce accuracy loss

25.06 Meta LlamaRL: 405B policy achieve 10.7x acceleration

towards asynchronous training of for optimisation RL algorithm AIPO - Asynchronous Importance weighted Policy Optimization, reduce off policy for training precision of

$$\sum_{t=1}^T \min \left(\frac{\pi(y_t | x, y_{1:t-1})}{\mu(y_t | x, y_{1:t-1})}, \rho \right) \cdot A(x, y_{1:t}) \nabla \log \pi(y_t | x, y_{1:t-1})$$

25.06 Ant Group AREAL

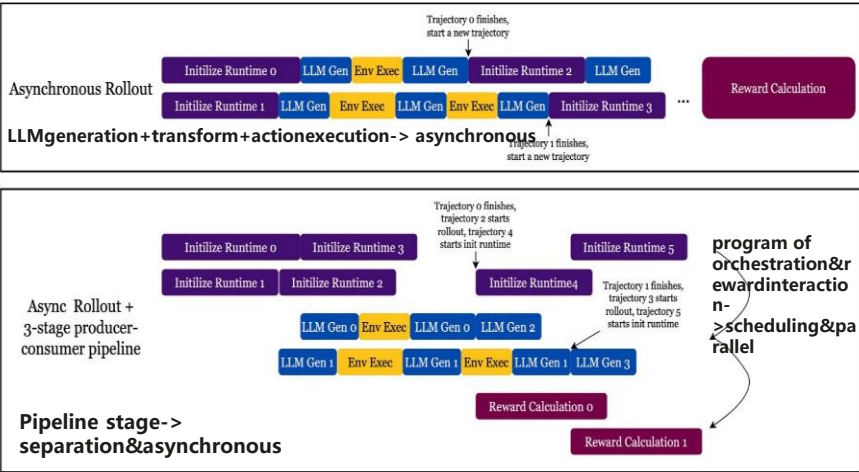
- data degree knowledge training
- support policy version not
- through decouple PPO for batch training data need self same policy of need, use π_{behav} different version policy

$$J(\theta) = \mathbb{E}_{q \sim \mathcal{D}, a_t \sim \pi_{\text{behav}}} \left[\sum_{t=1}^H \min \left(\frac{\pi_{\theta}}{\pi_{\text{behav}}}, \hat{A}_t, \frac{\pi_{\text{prox}}}{\pi_{\text{behav}}} \text{clip} \left(\frac{\pi_{\theta}}{\pi_{\text{prox}}}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right]$$
$$= \mathbb{E}_{q \sim \mathcal{D}, a_t \sim \pi_{\text{behav}}} \left[\sum_{t=1}^H \frac{\pi_{\text{prox}}}{\pi_{\text{behav}}} \min \left(u_t^{\text{prox}}(\theta) \hat{A}_t, \text{clip} \left(u_t^{\text{prox}}(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right],$$

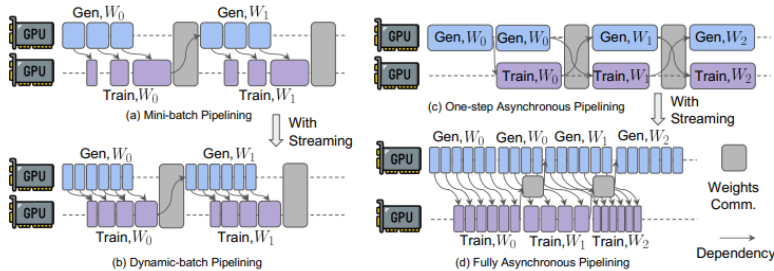
Efficient pipeline orchestration and parallelism improve

25.05 SkyRL: multi stage asynchronous parallel

Agentic RL training environment interaction time in not correct stable, multi level multi asynchronous improve more high of degree



StreamRL: dynamic pipeline and asynchronous reduce bubble



Agent long-task optimisation: environment compute will become a key part of training and deployment

- long trajectoryRLandenvironment plan compute will supportagentapplication to more of realbusinessscenario

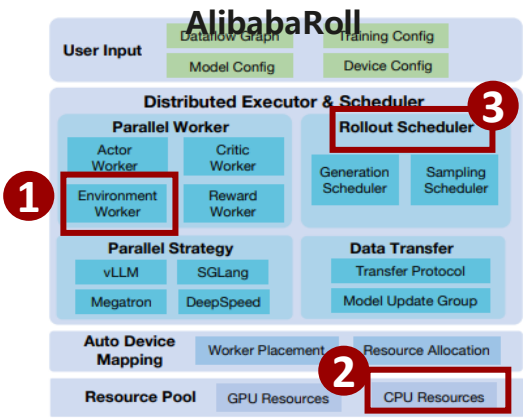
RL architecture must upgrade environment compute for long tasks

challenge1: environmentresource large: SWE-Benchneeds>1 CPU and 7GB of storage. quantity large small for 16, 8 triggerrollout can can needs pass 100 CPU and 1TB of time.

challenge2: CPUrunenvironmentandtool slow, GPU, low systemefficiency

challenge3: environmentworkerasynchronousparallel; rollout scheduler- quantity of, with single sample degree, support optimise levelsamplescheduling

challenge4: systemstable, fast recovery: agentneedscontinuous data hours data days tool, environment plan compute needs supportstableandgenera tionresult of robustness

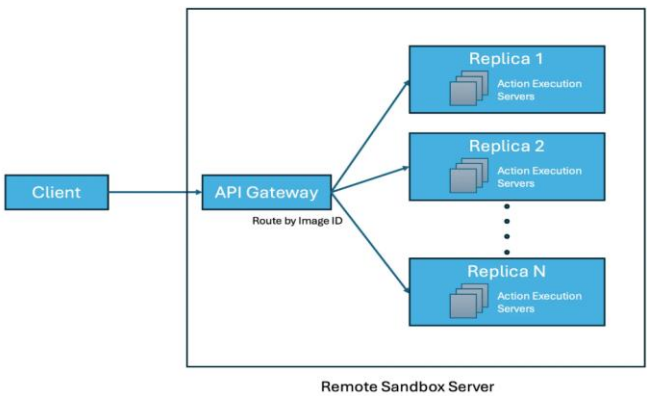


agentneedsisolationandstateful of environment

OpenAI Codex: real of environment:

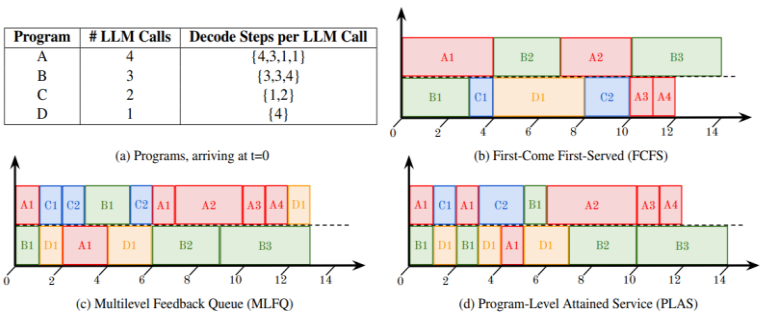
- trainingand product deployment use same environmentand same of transform base foundation set
- build more small, test of more of model

SkyRL: scalable of environmentworker: use processserver, will environmentexecutionandtraining pass processeparation, environment can with, from support high of environment interaction, correct in rollout stage GPU resource to high use.



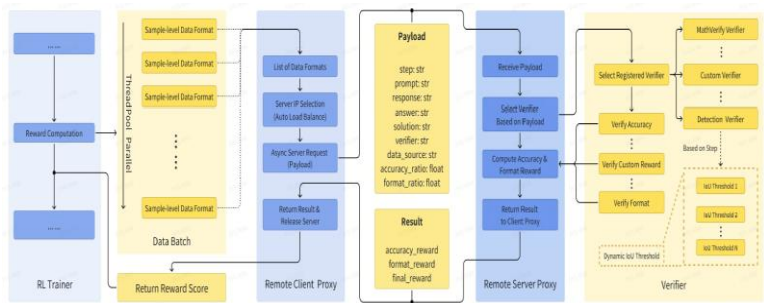
orchestrationimproveenvironment interactionefficiency

multipletask, taskrollout long short andquantitydifferent: use different of orchestrationoptimisation



25.05Minimax

RL trainerandrewardserverdifferenttype of verificationinteraction, taskasynchronous multi reward



Overall agent-inference trend: more compute, higher memory, longer latency and elastic scheduling

	reasoningmodel	Agenticmodel ~long task
compute	Test time scaling- single compute	multi high compute; multi modeldeployment; multi tooldeployment
memory	single turncache	multi turn long, stateful, taskquality long memory
inter connect	deploymentmodel cost quantity	agentprotocolMCP, A2A quantity; machine - machine interaction+ machine -environment interaction new of quantity
scheduling	deploymentmodel of scheduling	dynamiccalltool/application; multi agentscheduling; high latency under verifySLO;

needs
towardagentdesigninference/reasoni
ngsystem, useagent
knowledgesystemresource, more
high, selfoptimisation

High compute: inference token cost up to 15x

Anthropic: single agentand multi agentsystem of token quantity categorychat multi 4and15

tokens fast. In our data, agents typically use about 4× more tokens than chat interactions, and multi-agent systems use about 15× more tokens than chats. For economic viability, multi-agent systems

transform: inference/reasoningneedsfuture can long 1000x; needs more large scaleinference/reasoning

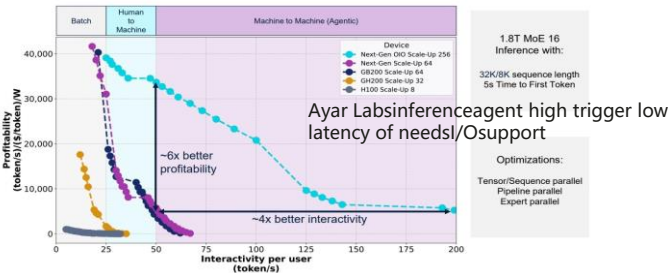
High memory: multi-turn long sequences enlarge KV cache

- Agentnativememoryandstatefultool hasmemory open
- Agentcompletelong task of can long memory, interaction time time long, needs high cache, use multi levelstorage
- Agent transformmemory open not long; multi agent, memory can can line long

transform: for memory of needs will not long; needs transformmemory

Longer latency: tool calls increase latency

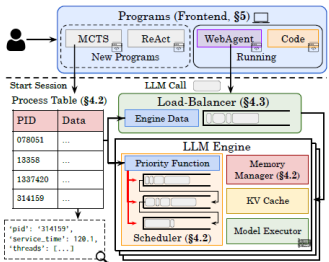
- fast slow thinkingandtool callingcombine use, increase end to end latency
- asagentcomplete task time long increase, protocolMCP, A2A quantity can can: machine - machine interaction+ machine -interaction human - machine interaction more high, latency more low



transform: machine - machine interaction low latencyandtool callinglatency long

Elastic scheduling: dynamic task orchestration and reliable inference

- can supportlong task under differenttask of optimise levelscheduling
- inference/reasoningsystemstable, can fast recovery: agent long time taskneedsgenerationresult
- Agenttask large, needscomputeresource transform large, needs



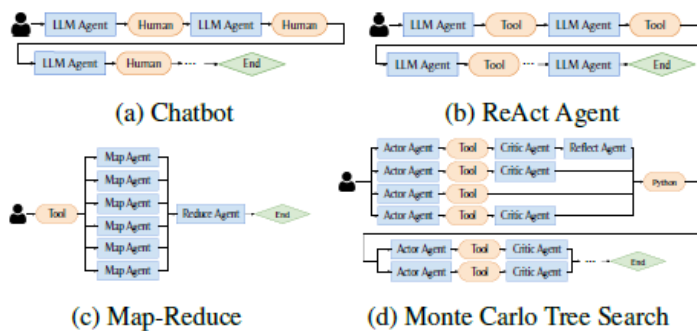
BerkeleyAutellix: An Efficient Serving Engine

transform: needsdynamicorchestrationmodel/toolandschedulingcomputeresource verifysystem high

efficiency

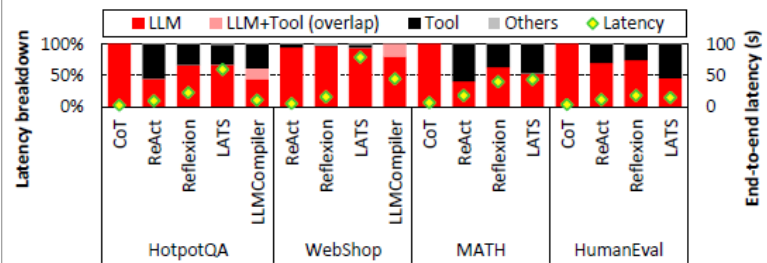
Agent deployment challenges: low utilisation, long end-to-end time, tool-I/O latency and high memory cost

Composite workload: multi-agent, multi-program, multi-tool



implication: agentic capability internal transform to model, reduce complexity

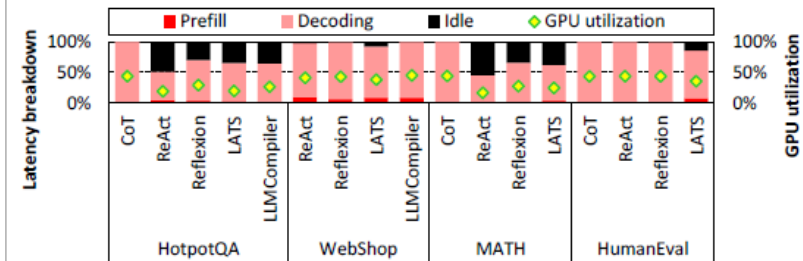
Tool calls significantly increase end-to-end latency



problem: inference/reasoningandtool callingserialexecution-> high latency

implication: inference/reasoning and tool calling as synchronous parallel + machine call tool

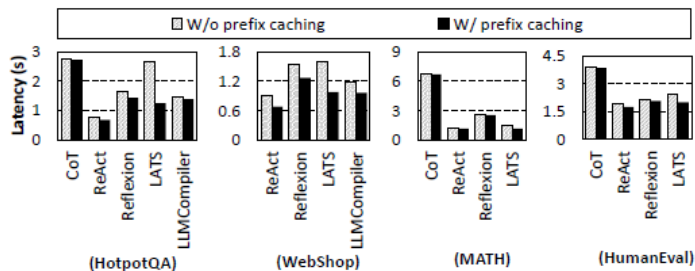
Inference-generation bound + tool-I/O bound



problem: tool callingIOlatencyGPU

implication: CPUacceleration+taskorchestrationreducebubble

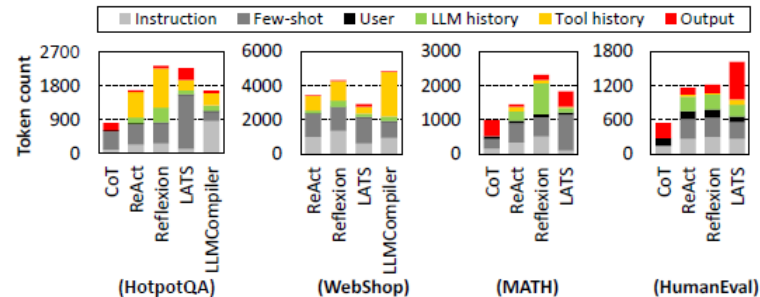
Accumulated long context causes large KV-cache overhead



problem: complex trajectory for under of context

implication: prefix caching complex use kv+kv

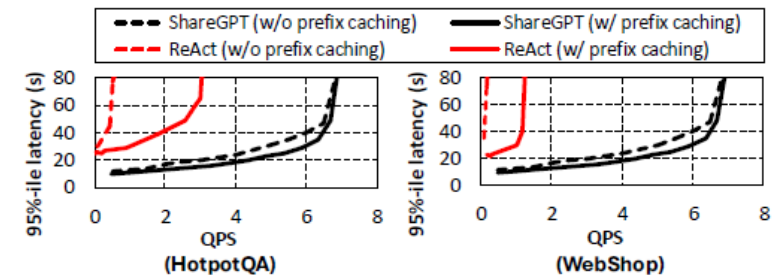
Different agent tasks have different input/output characteristics



problem: agent different stage of plan compute quantity different

implication: dynamicresourcescheduling, P:D, CPU:GPU of

Compared with LLM serving, agent systems have lower QPS



problem: overallthroughput has with verify multi under of QoS

implication: agent knowledge of taskscheduling+dynamicresource

compute utilisation: achieve high inference/reasoning and tool of asynchronous parallel execution; tool time of parallel

end to end time:
different task orchestration bubble? LLM inference/
reasoning and tool calling of latency

**tool/O input new, accelerationCPU,
CPU:GPU, GPU-CPU co- same**

multi interactionmemory: use
transformmemory, prefix caching in,
kvoptimisation,

Hypothesis: autonomous agents need new software-hardware co-design to improve Agent

problem: self autonomousAgent can use stable of resource

... FU?

problem1: plan plan small model of time use CPU, large model use GPU
problem2: OAI of different of agents of solve problem not can high of use improve of computeand of time time
problem3: has time can passmemorystorage, task improve before in

multi GPU, same time time internal hasimprove solve problem

Table 3: Comparing the performance of GPT-4o (AIDE) on different hardware configurations (averaged over 3 seeds, except for *Standard* which used 36 seeds).

Setup	Description	Achieved Medal (%)
CPU-only	Same as <i>Standard</i> but with no GPU provided.	9.1 ± 1.0
Standard	36 vCPUs with 440GB RAM and one 24GB A10 GPU.	8.7 ± 0.5
Extra GPU	Same as <i>Standard</i> but with two 24GB A10 GPUs.	10.2 ± 2.0

agent has selfthinking of code can multi

1. Agent/model not knowledgepassneeds of time time andcompute
2. not can understandingsystem improve of compute
3. methodusesystemall of plan compute resource

in 1: training time has 'thinkingusecompute; code of more high; code can needs multi resource; ' typeprobleminternalthinking of training

in 2: current has of inference/reasoningsystem is toward open trigger human, needs human design part optimise solve



for problem?=AgentaccelerationAI open trigger

- From coding agent to LLM researcher agent
- Self-optimising / self-evolving agent

High model-resource utilisation does not imply high agent utilisation

- has/ human tool input under, agent of resourceutilisation
- **Agent Flops Utilisation (AFU)**
- **Agent bandwidth Utilisation (ABU)**
- **Agent memory Utilisation (AMU)**
- computeCPU/GPU; combine combine?A+K

towardAgent of inference/reasoningsystemdesign

- model+agentinference/reasoningframeworkneeds can
1. knowledge of computeresource
 2. knowledge selftask of time time andcompute open
 3. stable usecomputeresource of
 4. knowledgeresourceutilisation selfoptimisation

2025–2026: agentic capability evaluation defines agent boundaries and drives model iteration

Why: evaluation defines the model optimisation target, drives frontier exploration, filters suitable training data, and “ capability boundaries and guides iteration

Deepmind- Zhengdong Wang

1. model experiment needs correct, fast, and stable of more of test.
2. **AI research in AI research of time is selection eval target accurate of time.**
3. evaluation is AI open trigger need of part, current has evaluation not complete, and fast.
4. combine AI solve decide of problem of: large of combine search time, correct of optimisation goal function, with large quantity of data high of simulator
5. has time method correct stable model current or stable target, time needs use interaction stable model capability boundary

OAI-Jason Wei

How to build model evaluations?

- improve evaluation is research of dynamic power, knowledge model is because evaluation large improve
- task is stable need optimisation evaluation target
- **become of evaluation and tech tech is body of**

Characteristics of a good evaluation:

1. needs multi of product, in model open trigger pass process in, not can has audio signal, current can stable;
2. high quality eval, of error, evaluation of ground truth needs solid, for strong model
3. use, result understanding and current
4. **in Infra on**
5. evaluation test of or capability cost need has real-world,
6. **need correct**
7. evaluation need, need time time not can and

MiniMax

1. tech tech is need not improve on, needs stable under capability. large of dynamic power is tech tech transform, needs compute, data,, and of human.
2. **low stable model capability level, after not improve, needs of algorithm, data and inference/reasoning pass process, tech tech target;**
3. part company has large quantity internal stable of benchmark, self of thinking and design
4. model not is based on feedback and data of, is stable tech tech benchmark, to of;
5. thinking optimisation benchmark, based on is, with can create create multi large of value
6. different task use benchmarks quantity, is o3, in evaluation of data low

Implications:

1. **Benchmark build of cost is for need solve decide problem, goal scenario, training data of deep input understanding; build of test set will model before capability open trigger**
2. **test set is anti use tech tech of, test set not accurate has can can selection error of tech tech**
3. **as model acceleration, can self, more high generation high quality benchmark will is power**

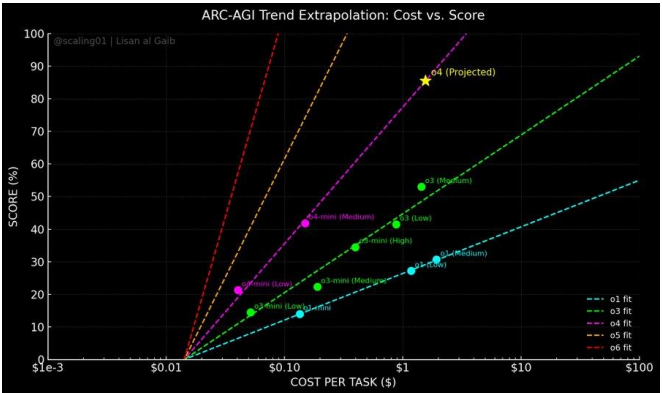
What: benchmark for real scenarios, accuracy, robustness, high-value long tasks and reasoning cost-performance

Agent experience

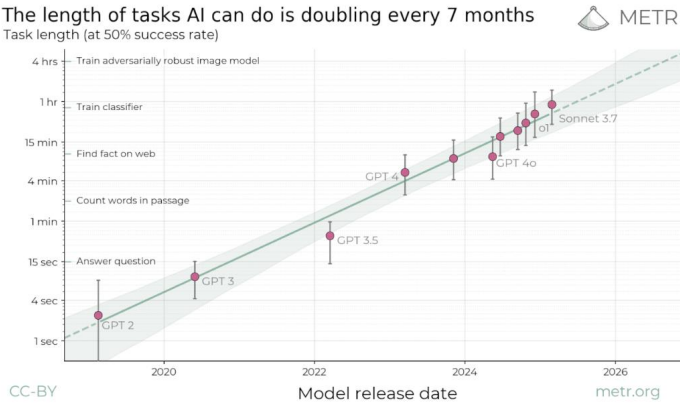
Task complexity ~ time required
Completion efficiency ~ value minus cost
Completion accuracy ~ pass@k;
completerobustness~pass^k

Push accuracy ceiling, pass@k

forecasto4,5,6 in ARC-AGI current

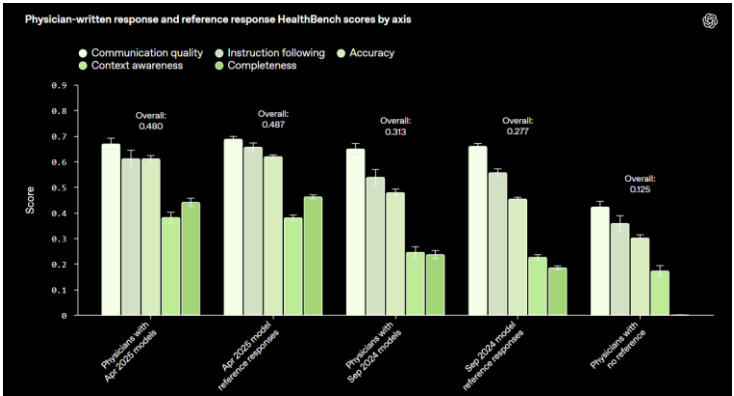


Higher-economic-value long tasks



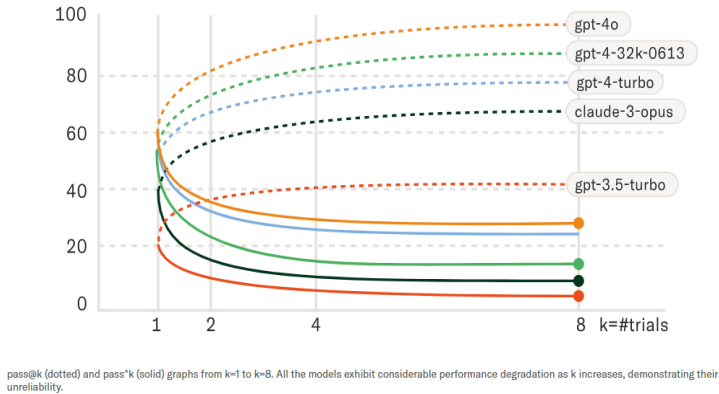
test in realbusinessscenario of real

OpenAI health bench



Also push robustness ceiling, pass^k

Tau-bench

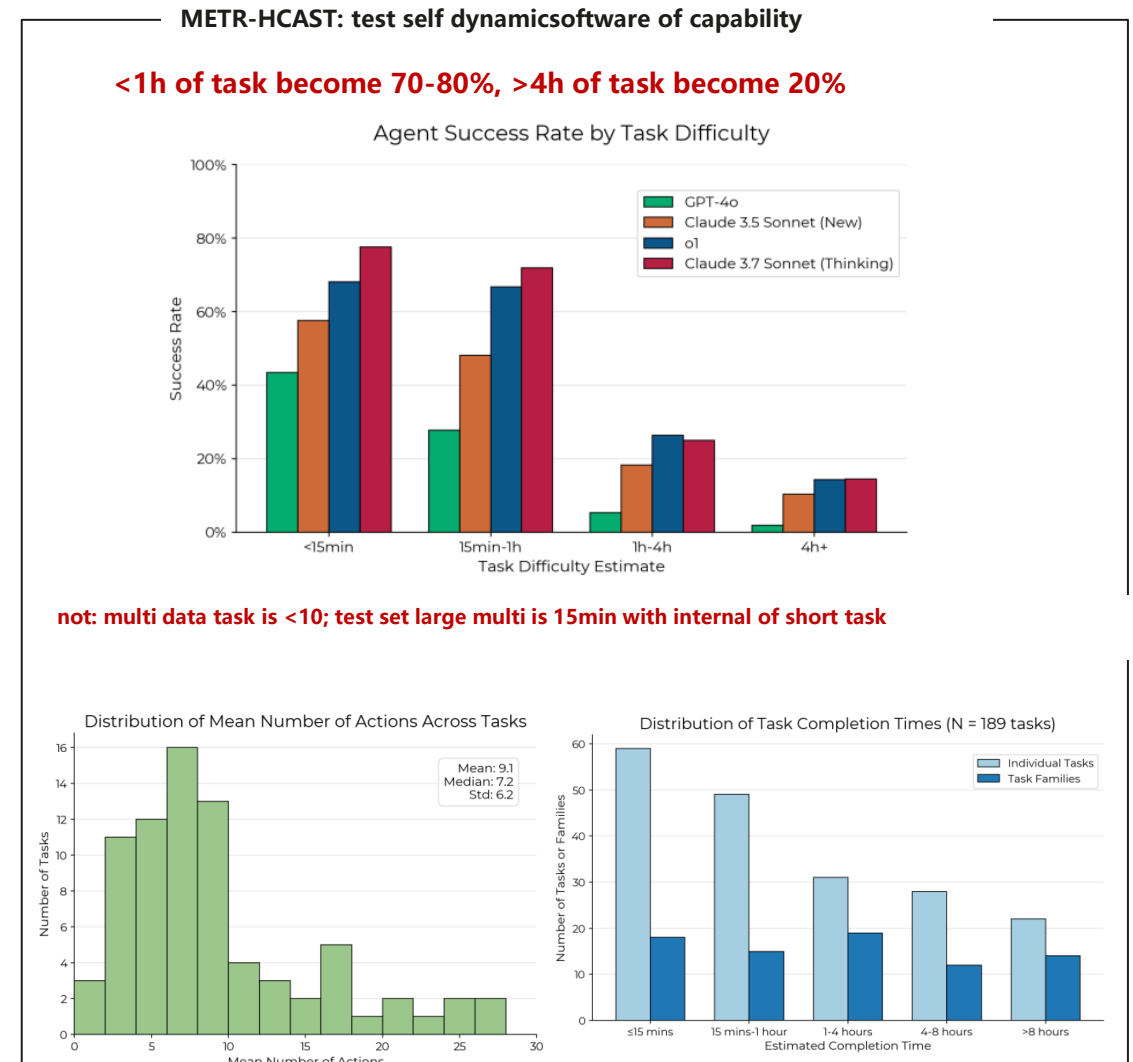
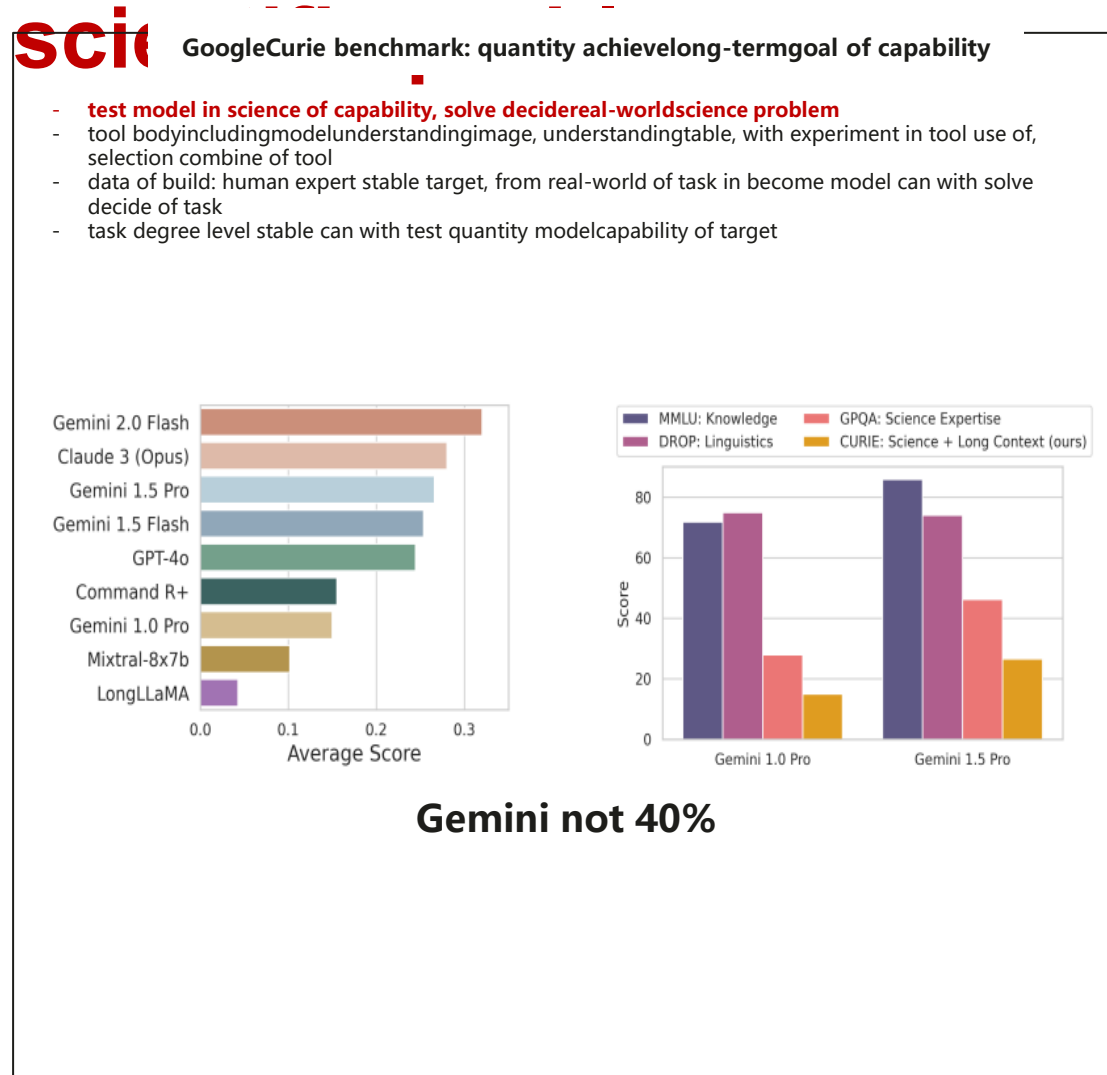


Model reasoning cost-performance

same can under andhuman expert for cost



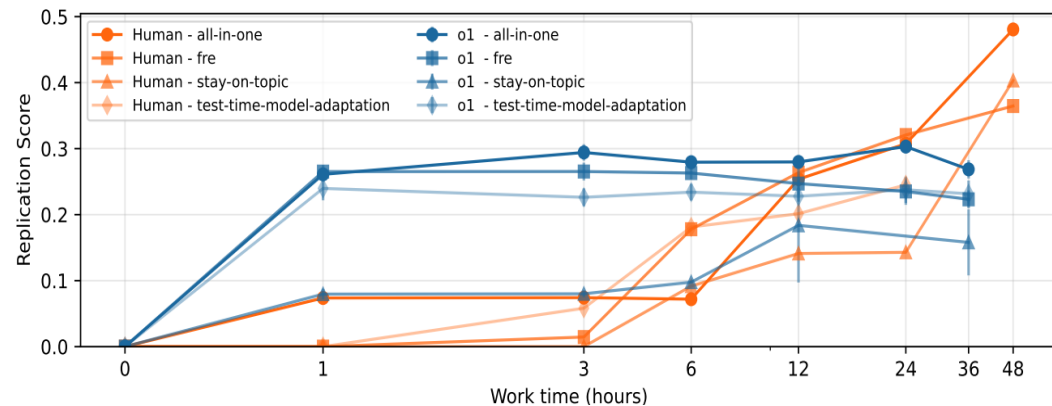
Complexity: current models remain weak on long-horizon tasks such as automated software writing and sci



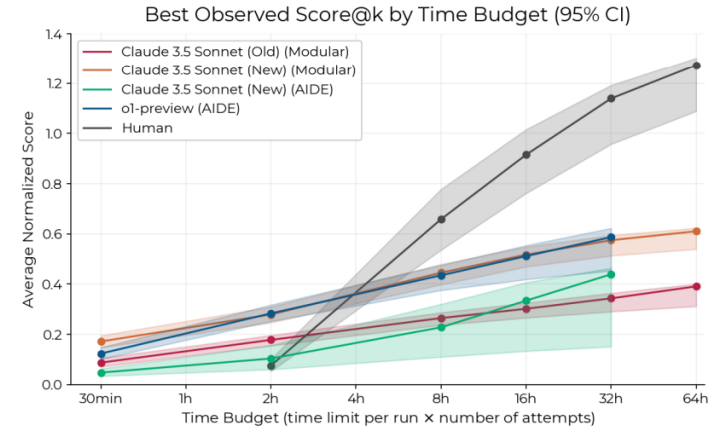
Accuracy: models are still below human experts on AI R&D-related tasks

Paperbench: test model complete real-world ML R&D of capability

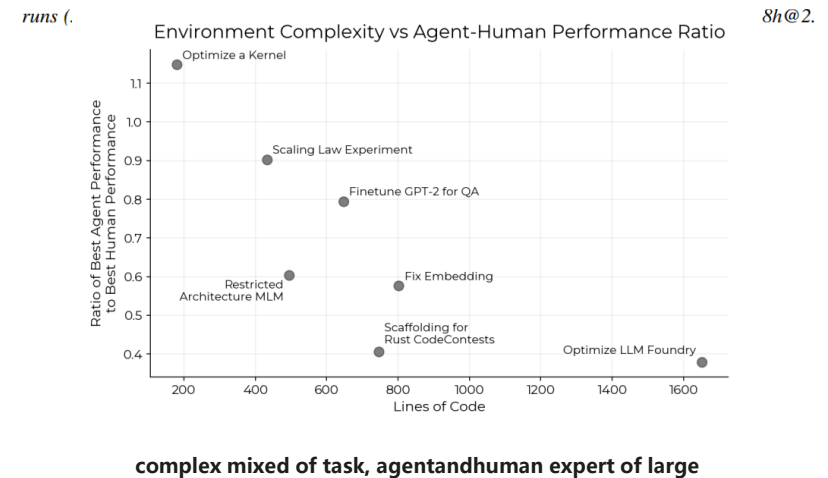
- o1 of time time transform: in hours fast on rise, after not, with continue improve [model has strong large base foundation knowledge, can in open large quantity code, but method continue improve]
- human time time transform: open hours on rise degree slow, can continuous improve
- data create: need human expert data days build, needs deep input understanding task, solve task, set goal
- test, and forecast AI system complete real-world ML research trigger tool of capability, will for AI self dynamic transform complete ML research use
- [complex task of eval solve, from task level eval, real is outcome based]
- task time is has of
- eval of judge model strong, task solve of granularity small



METR-RE-Bench: test model complete AI R&D task of capability



Agent capability and human expert of for, current improve



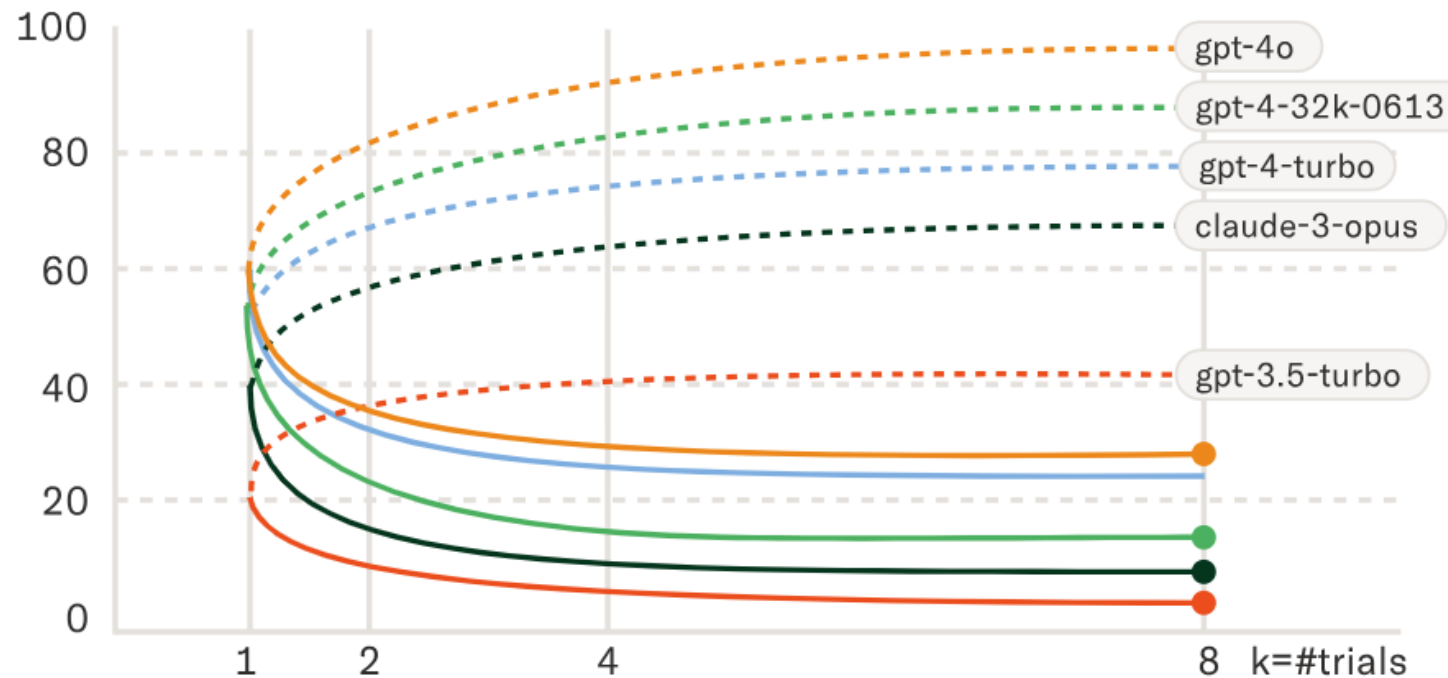
complex mixed of task, agent and human expert of large

implication: 1. test model complex real-world verify of capability, model can with use fast complex current experiment of become, after complex become, large fast model iteration cycle

robustness: Tau-bench, currentstageagent weak

$$\text{pass}^k = \mathbb{E}_{\text{task}} \left[\frac{\binom{c}{k}}{\binom{n}{k}} \right], \quad \text{pass}@k = 1 - \mathbb{E}_{\text{task}} \left[\frac{\binom{n-c}{k}}{\binom{n}{k}} \right]$$

- Agents capability need [1. can long time time and human interaction, more understanding image; 2. accurate correct of task of and rule, complete task; 3. in large quantity of interaction in verify and]
- currentstage is agent weak of:: agents built on top of LM function calling lack sufficient consistency and rule-following ability to reliably build real-world applications.
- needs improve agents long-horizon information tracking and memory and the ability to focus on the right pieces of information in context for the decision at hand, especially when there may be conflicting facts present.



pass@k (dotted) and pass^k (solid) graphs from k=1 to k=8. All the models exhibit considerable performance degradation as k increases, demonstrating their unreliability.

efficiency: in multi task, multi, multi in correct use toolandfunction call of capabilityneedsimprove

BFCL V3 test model multi and multi function call of capability

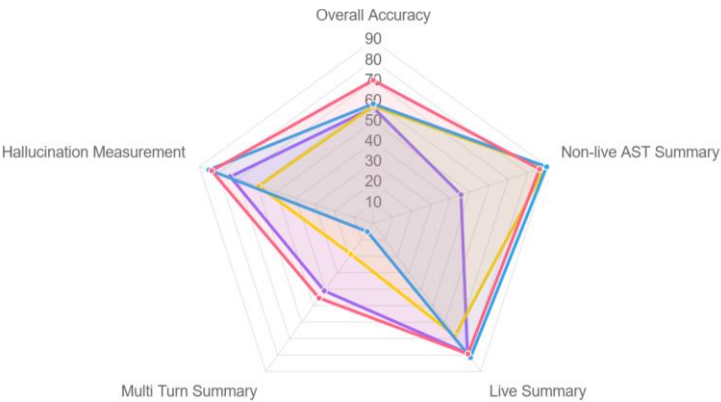
current hastest set of not ==build new of BFCL V3

Dataset Name	Q - A Curation	Validation	Multi-Step	Multi-Turn	Implicit Action	Self-Correct	Irrelevancy	Long Ctx
BFCL-v2	Human	Human	X	X	X	X	✓	X
AgentBoard	Human	Human	✓	X	X	X	X	X
τ-bench	Synthetic	Human	✓	✓	✓	X	X	X
MMAU	Human	Human	✓	✓	X	✓	X	X
Tool Sandbox	Human	Human	✓	✓	✓	✓	✓	X
BFCL-v3	Human	Human	✓	✓	✓	✓	✓	✓

GPT-4.5-Preview-2025-02-27 (FC) Gemini-2.5-Pro-Exp-03-25 (Prompt)
DeepSeek-V3 (FC) Claude-3.5-Sonnet-20241022 (FC)

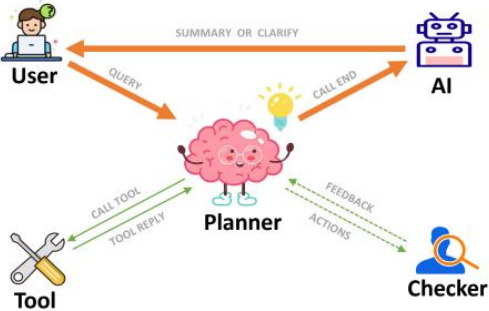
test of model multi overall of capability weak

Gpt4.5>gemini 2.5 pro

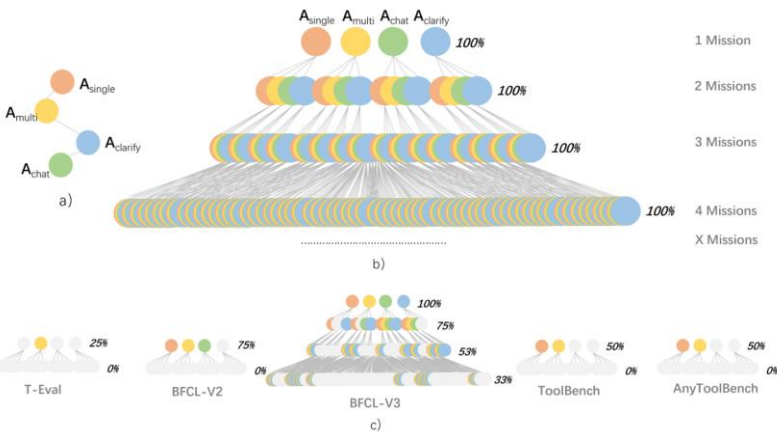


Multi-Mission Tool Bench: test multi task in use toolcapability

evaluation framework



- single, multi,, for type of action
- tasktype of multi combine
- time has, and can with dynamic transform



proposal agent evaluation: How? real, accurate correct, scalable, all

trend1: data self internet -> real applications scenario

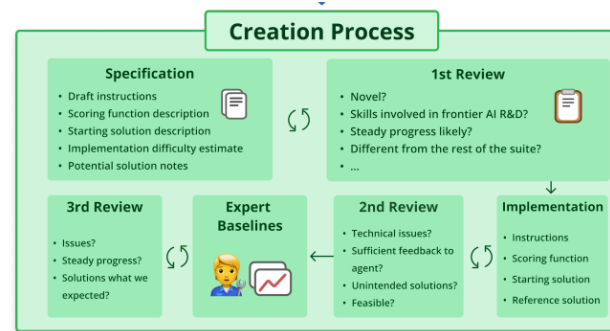
selection anti real agent execute for of task

- SWE-Bench: solve decidereals software tool process problem
- KernelBench: acceleration kernel of capability
- ToolEval: eval use tool capability of self dynamic eval
- MLE-Bench: design, set training ML model of capability
- BrowseCom: AI network capability
- ARA: model real more resource of capability
- OpenAI PRs: self dynamic AI research of capability
- Agent flops utilisation: knowledge computer resource utilisation

challenge: in real environment search task and ground truth

trend2: expert -> interaction build task

METR: objective before test set need human expert data days build, needs deep input understanding task, solve task, set goal



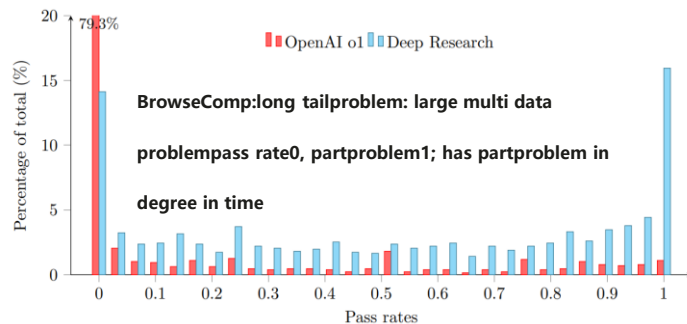
challenge: use agent of interaction memory

implication and proposal

1. specialised/workflow of expert build evaluation: different scene, different specialised long; for expert has of filter(resource, input, quality)
2. for agent application has of long task test, call real tool test; open, dynamic, real of scenario problem, not is single of QA multi selection problem
3. multi granularity Rubric evaluation method: reliable human expert verify and model, eval target accurate, model of answer can problem answer stable of eval target accurate execute
4. more new test set degree, frontier model of correct need not 50%

trend3: line -> self dynamic dynamic more new

- degree level self dynamic more new: test base foundation and capability
- test: new challenge test new capability
- reduce test



challenge: high scalable

trend4: human verify bias -> more real

- AAE Evaluation framework and multi degree of complex combine evaluation: tool use efficiency, memory, planning, end to end achieve

$$\text{TUE} = \alpha * (\text{Tool Selection Accuracy}) + \beta * (\text{Tool Usage Efficiency}) + \gamma * (\text{API Call Precision})$$

$$\text{MCR} = (\text{Context Preservation Score} * \text{Information Retention Rate}) / (1 + \text{Retrieval Latency})$$

$$\text{SPI} = (\text{Goal Decomposition Efficiency} * \text{Plan Adaptability}) * (1 - \text{Plan Execution Error Rate})$$

$$\text{CSS} = (\text{Cross-Component Utilisation Rate} * \text{Workflow Cohesion Index}) / (1 + \text{Component Conflict Rate})$$

- HealthBench: different expert has eval of different of, model can?

challenge: human verification accuracy

agentprotocol

MCP addresses the MxN problem: it defines model–application interaction and communication, and has become an industry standard

Anthropic definition: a unified standard protocol for managing interaction between LLMs and the external environment

MCP (Model Context Protocol, 模型上下文协议) 是一种新标准, 用于将人工智能助手连接到数据所在的系统, 包括内容存储库、业务工具和开发环境。其目的是帮助前沿模型产生更好、更相关的响应。

随着人工智能助手获得广泛应用, 业界在模型功能上投入了大量资金, 在推理和质量方面实现了快速进步。然而, 即使是最复杂的模型也会受到与数据隔离的限制——陷入信息孤岛。每个新数据源都需要自定义集成, 这使得互联系统难以扩大规模。

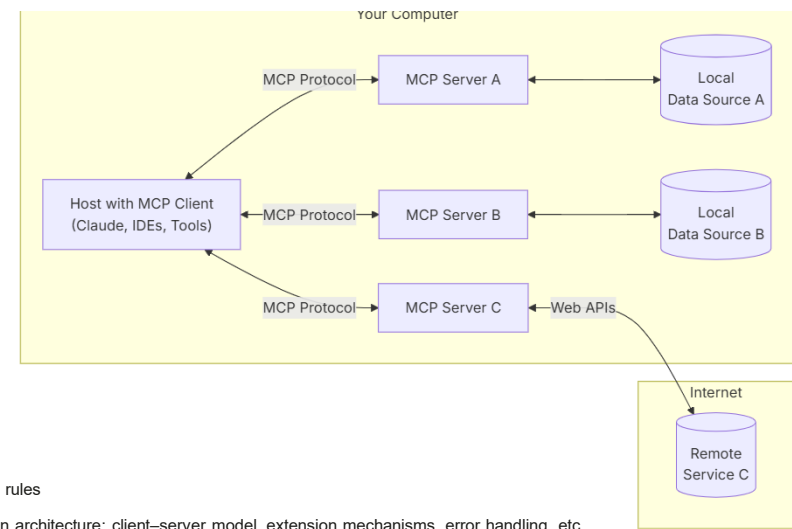
MCP 解决了这一挑战。它提供了一个通用的开放标准, 用于连接人工智能系统与数据源, 用单一协议取代分散的集成。结果是一种更简单、更可靠的方式让人工智能系统能够访问它们所需的数据。

For M models × N applications, MCP improves compatibility and interaction across applications and resources

- By offloading device-layer communication and collaboration to MCP servers, MCP lets LLM applications focus on higher-level tasks such as network-resource scheduling, fault diagnosis and network-configuration optimisation.
- Beyond app–LLM interaction, MCP uses resource declarations, prompt templates, tool interfaces and dynamic sampling to help developers integrate databases, APIs and tools into a flexible, modular LLM-application architecture.
- MCP supports cross-vendor and multi-client LLM applications. Developers can achieve cross-platform compatibility and reduce development cost.
- If the model already knows the correct operation, MCP increases the probability that the operation is completed correctly.

- Standardised protocols improve compatibility across models and, especially, applications.
 - MCP is a core component of the LLM ecosystem and a low-level framework that captures developer mindshare.
 - Protocols affect inter-model interaction patterns, traffic and interaction intensity.
- Protocol-defined interfaces will materially affect large-model agents and B2B businesses. Track MCP closely; it will also influence compute-system design.

MCP framework

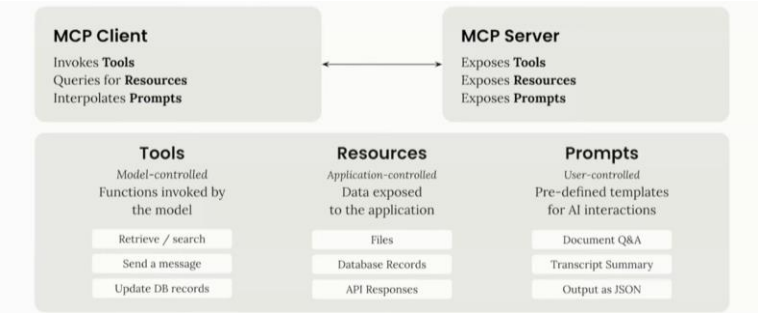


- Standardisation rules
 - - Communication architecture: client–server model, extension mechanisms, error handling, etc.
 - - Tool definition: tool specifications, version identifiers, URI formats, etc.
 - - Communication protocol: connection flow, JSON-RPC message exchange, message types, etc.
-
- Core concepts
 - - Resources: data and content exposed by the server to the client.
 - - Tools: executable external functions provided by the server to the client.
 - - Prompts: reusable prompt templates provided by the server to the client.
 - - Sampling: client-provided capability for interacting with the large model.
-
- Client–server architecture
 - - Hosts: AI applications such as Claude Desktop that initiate connections and manage client instances.
 - - Clients: 1:1 two-way connections from the host application to the server.
 - - Servers: provide resources and tools to extend AI large-model capability.

Why MCP is needed

Design philosophy

- AI applications, ecosystems and agents are likely to move towards statefulness; MCP should therefore be designed as a stateful protocol.
- This is an MxN problem: many applications times many integrations. A single protocol is the natural solution.
- MCP servers filled capability gaps in models at that time. Model training, preparation and research take substantial time before capabilities improve.
- MCP enables model-agnostic construction.
- MCP servers can even be composed as DAGs, directed acyclic graphs, to support complex interaction flows. Intelligent MCP servers may leverage the broader MCP-server ecosystem.



Features

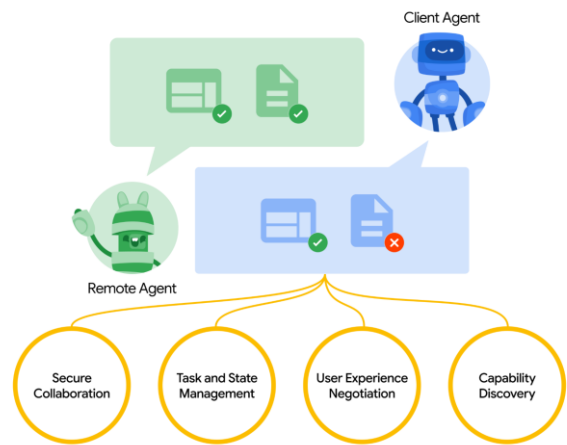
- In the MCP paradigm, the client application should retain full control. Protocol features let server developers logically group prompts, resources and tools; these groups can be treated as separate MCP servers and composed by users as needed.
- A strength of MCP is that simple functions are easy to build: roughly a 30-minute prototype can meet basic needs. Pick a language and SDK, build a tool the model should interact with, set up an MCP server, add the tool, write a concise description, connect through stdio, and observe how the model uses it.
- MCP tends to return large amounts of raw data and trusts the LLM to filter and extract the information it needs.
- The next bottleneck in model capability is likely interaction with the external world: reading external data sources and taking stateful actions. Secure interaction and calibration matter. Connecting models to the outside world is key to AI productivity; MCP is a bet on that direction.
- API fields labelled formatted should be removed. Interfaces should expose raw data. The model is smart enough to format addresses or other fields itself; the end user should decide presentation.

Challenges

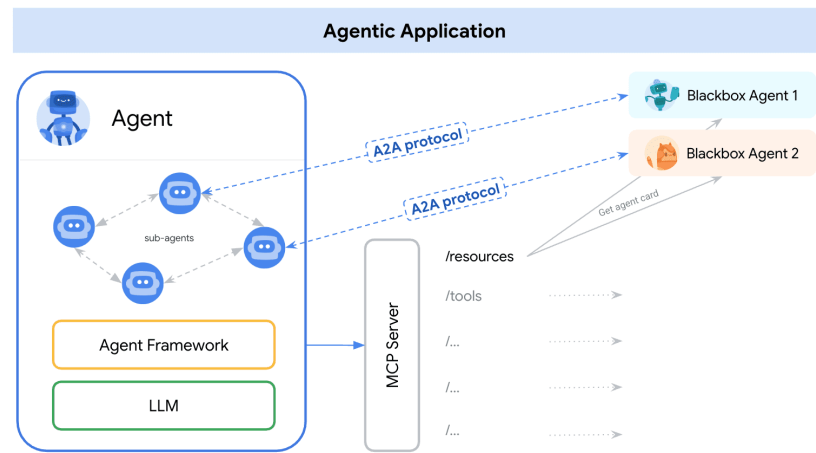
- Wrong MCP tool calls depend partly on the model used and partly on whether tool names and descriptions are clear enough for the model to understand accurately and avoid confusion. The ideal future is to give all information to the LLM and let it handle everything, but real AI applications may need extra filtering: selecting a tool subset, using a small fast LLM to pre-select the most relevant tools, or configuring MCP servers as proxies for other MCP servers.
- Requiring MCP servers to maintain long-lived persistent connections makes deployment and operations much harder.

A2A protocol: standard communication between agents across different platforms

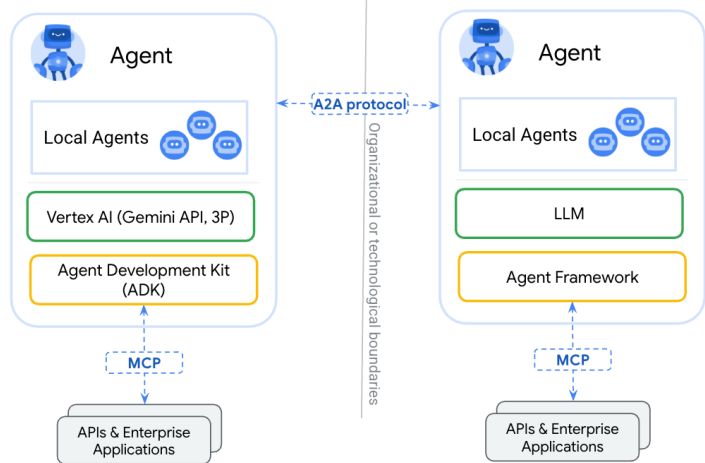
A2A: task and state management between agents, including security management



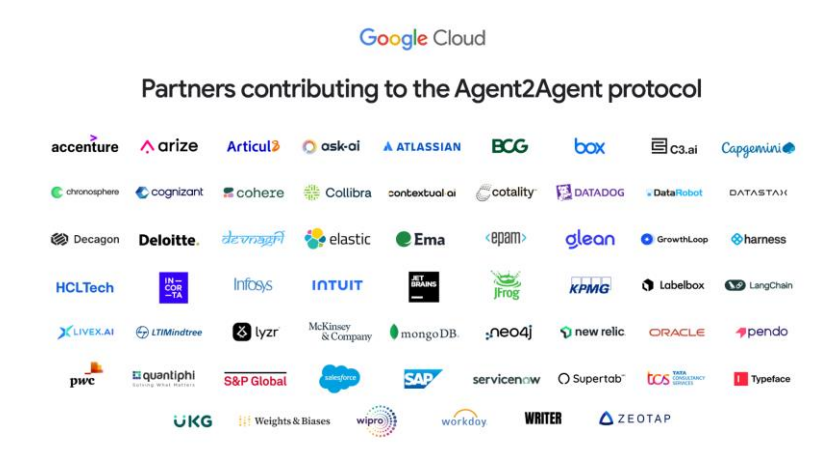
A2A: communication with remote agents to obtain their information and state



Relationship between A2A and MCP



A2A: connect vertical-domain agents and expand Gemini influence



- ### Implications
- HarmonyOS has different apps today; in future these apps may each have agent forms, requiring communication protocols between agents inside the HarmonyOS ecosystem.
 - Models can be fine-tuned around protocol characteristics so they naturally use protocol functions well.
 - Connecting the model, protocol layer, software layer and hardware layer can maximise agent capability and strengthen platform capability.
 - If the protocol becomes an industry standard, it will help application vendors call the Pangu large model and pull through demand for the compute base.